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A parallel finite element solver for parabolic equations on quantum graphs

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Abstract

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Introduction

1 Analytical and numerical aspects of quantum graphs

Quantum graphs have attracted significant interest as a tool for modeling phenomena involving a wide range of physical systems: from wave propagation in photonic crystals and thin waveguides to molecular dynamics and signal transmission in social, financial networks and neural architectures in the context of neurosciences. One can find a collection of papers on quantum graphs in [1]. Together with their potential interdisciplinarity, they gained popularity also because they naturally arise as reduced-order models.
Here we provide the essential theoretical background on quantum graphs needed for the computational development of this project. We emphasize that the material covered here is by no means exhaustive; a comprehensive introduction to quantum graphs can be found in [2]. Therefore, the purpose of this section is to introduce the reader to the class of mathematical models known as *quantum graphs*, first from an analytical perspective and then by describing the numerical methods used to study them.

1.1 Basic definitions

Let us begin with an intuitive definition. A *quantum graph* is a collection of one-dimensional intervals connected at their endpoints to form a network. The term "quantum" refers to the fact that the graph is equipped with a Hamiltonian operator that acts on the functions supported in these intervals, coupled by suitable boundary conditions at the vertices [3]. We already provided the main idea behind quantum graphs, but for a more rigorous formulation, we need the following Definitions:

Definition 1.1 (directed graphs)**.** *A **directed graph** is a pair of sets $\Gamma = (\mathcal{V}, \mathcal{A})$, where:*

- $\mathcal{V} = \{v_i\}_{i=1}^N$ *is a set of **vertices** (or nodes);*
- $\mathcal{A} \subseteq \mathcal{V} \times \mathcal{V}$ *is a set of ordered pairs of vertices, called **arcs**.*
- *Given an arc $(v_i, v_j) \in \mathcal{A}$, it is said to be* outgoing *from v_i and* incoming *to v_j. Two nodes v_i and v_j are said to be adjacent if $(v_i, v_j) \in \mathcal{A}$.*

Definition 1.2 (undirected graphs)**.** *An **undirected graph** is a pair of sets $\Gamma = (\mathcal{V}, \mathcal{E})$, where:*

- $\mathcal{V} = \{v_i\}_{i=1}^N$ *is a set of **vertices** (or nodes);*
- $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ *is a set of unordered pairs of vertices, called **edges**.*
- *An edge connecting two vertices v_i, v_j is written $\{v_i, v_j\}$ with $i < j$. Two nodes v_i and v_j are said to be adjacent if $\{v_i, v_j\} \in \mathcal{E}$.*

Definition 1.3 (combinatorial and metric graphs)**.** *Definition 1.1 and Definition 1.2 present arcs and edges as binary relations between pairs of vertices. Two nodes are either linked together or not: we don't know anything about the "strenght" of the connection. For this reason, the graphs that we presented are called **combinatorial graphs**. One can choose a measure for the length of such connections. A combinatorial graph endowed with geometry is a **metric graph**.*

Now we present some useful matrices used for the description of combinatorial (and metric) graphs.

Definition 1.4 (adjacency matrix)**.** *The **adjacency matrix** of a graph Γ with N vertices is the symmetric $N \times N$ matrix $A \in \{0,1\}^{N \times N}$, defined by*

$$A_{ij} = \begin{cases} 1 & \textit{if vertices } v_i, v_j \textit{ are adjacent,} \\ 0 & \textit{otherwise.} \end{cases}$$

Definition 1.5 (incidence matrix)**.** *The* ***incidence matrix*** *of a directed graph* $\Gamma = (\mathcal{V}, \mathcal{A})$ *with* N *vertices and* M *arcs is the* $N \times M$ *matrix* E*, whose column corresponding to arc* (v_i, v_j) *has exactly two nonzero entries:* $+1$ *in row* i *and* -1 *in row* j*. The same holds for an undirected graph with edges* $\{v_i, v_j\}$*.*

Definition 1.6 (combinatorial Laplacian)**.** *The* ***combinatorial Laplacian*** *of* Γ *is the* $N \times N$ *symmetric positive semidefinite singular matrix*

$$L_\Gamma = EE^\top.$$

For the analysis of the properties of the combinatorial Laplacian, we refer to Appendix A.

Definition 1.7 (natural coordinates)**.** *Each edge* e *of a (undirected) metric graph can be associated to a finite interval* $(0, \ell_e) \subset \mathbb{R}$*, with* ***natural coordinate*** s_e*, so that* s_e *gives the position of a point along edge* e*. Note that assigning a coordinate requires a choice of direction: we take the increasing direction of* s_e *to coincide with the orientation fixed when defining the incidence matrix* E*. For* $e = \{v_i, v_j\}$*,* s_e *will therefore increase in the direction that goes from* v_i *to* v_j*, with* $i < j$*. The same definition holds for natural coordinates on arcs of a directed graph, with increasing direction coinciding with the one of the arc itself.*

The definitions of metric graphs and natural coordinates are very important to introduce functional spaces over Γ.

Definition 1.8 ($L^2(\Gamma)$)**.** *Let* $\Gamma = (\mathcal{V}, \mathcal{E})$ *be a metric graph. For each edge* $e \in \mathcal{E}$*, let*

$$L^2(e) := \left\{ f : e \to \mathbb{R} \;\middle|\; \int_e |f|^2 \, \mathrm{d}s < \infty \right\}.$$

The space of square-integrable functions on Γ *is defined as*

$$L^2(\Gamma) := \bigoplus_{e \in \mathcal{E}} L^2(e).$$

In other terms, $f \in L^2(\Gamma)$ *if and only if*

$$\|f\|^2_{L^2(\Gamma)} = \sum_{e \in \mathcal{E}} \|f\|^2_{L^2(e)} < \infty.$$

Definition 1.9 ($H^1(\Gamma)$)**.** *For each edge* $e \in \mathcal{E}$*, let*

$$H^1(e) := \left\{ f \in L^2(e) \;\middle|\; \int_e |f'(s)|^2 \, \mathrm{d}s < \infty \right\}.$$

One definition of $H^1(\Gamma)$ *is:*

$$H^1(\Gamma) := \left(\bigoplus_{e \in \mathcal{E}} H^1(e) \right) \cap C^0(\Gamma),$$

where $C^0(\Gamma)$ *is the space of continuous functions on* Γ*.*
In other terms, $f \in H^1(\Gamma)$ *if and only if* f *is continuous on* Γ *and*

$$\|f\|^2_{H^1(\mathcal{G})} = \sum_{e \in \mathcal{E}} \|f\|^2_{H^1(e)} < \infty.$$

The idea behind the definition of a quantum graph is that edges may be characterised not only by constant coefficients, but also by variable ones following physical laws, possibly described by differential equations. In particular, vertices can interact through such laws. To formalise this, we assign to the metric graph Γ an additional structure represented by a differential operator $\mathcal{H}$, called the *Hamiltonian*. In particular:

$$\mathcal{H}\, u(s) := -\frac{d^2 u(s)}{ds^2} + V(s)\, u(s), \tag{1}$$

where $V(s) \in L^\infty(\Gamma)$ is a given potential and derivatives are taken along the edge coordinate s. To pose a well-defined differential problem on Γ, we must supplement $\mathcal{H}$ with a set of vertex conditions $\mathcal{C}$.

Definition 1.10 (quantum graphs). *A* quantum graph *is a triple* $(\Gamma, \mathcal{H}, \mathcal{C})$*, where* $\Gamma = (\mathcal{V}, \mathcal{E})$ *is a metric graph,* $\mathcal{H}$ *is a differential operator acting edgewise on it and* $\mathcal{C}$ *is a set of vertex conditions imposed at* $\mathcal{V}$*.*

Examples of boundary conditions are the so called *Neumann-Kirchhoff conditions*.
The remainder of this report is devoted to the investigation of differential problems on quantum graphs. In particular, we will notice that also more complicated operators acting on Γ can be considered, as the one generating parabolic problems.

1.2 Finite element discretization

The classical finite element method can be employed to study quantum graphs. Let $\Gamma = (\mathcal{V}, \mathcal{E})$ be a metric graph, where each edge $e \in \mathcal{E}$ is identified with an interval $(0, \ell_e)$. For an edge $e = \{v_p, v_q\}$, we fix an orientation so that $s_e = 0$ corresponds to v_p and $s_e = \ell_e$ to v_q. On each edge e, we introduce a uniform partition into n_e subintervals of length $h_e = \ell_e/n_e$. This yields the nodes

$$\{s_j^e\}_{j=0}^{n_e}, \qquad s_j^e = jh_e,$$

with $s_0^e = 0$ and $s_{n_e}^e = \ell_e$, corresponding to the vertices v_p and v_q, respectively. The interior nodes are $\{s_j^e\}_{j=1}^{n_e-1}$. This construction induces a new graph, which we refer to as the *extended graph*. Its vertex set consists of the original vertices $\mathcal{V}$ together with all the interior nodes introduced by the discretization, while its edge set is given by the segments connecting consecutive nodes along each original edge. On this extended graph, we construct the standard hat basis functions $\{\phi_j^e\}_{j=1}^{n_e-1}$, associated with the interior nodes of each edge, together with the functions $\{\phi_v\}_{v\in\mathcal{V}}$, related to the original vertices. All these functions are piecewise linear and globally continuous on Γ. More precisely, each function ϕ_j^e is supported on the two elements adjacent to the node s_j^e, takes the value 1 at s_j^e, and vanishes at all other nodes. Similarly, for each vertex $v \in \mathcal{V}$, the function ϕ_v takes the value 1 at v and decreases linearly to zero along each edge incident to v, vanishing at the neighboring nodes. In practice, we first construct, on each edge, the usual hat functions of the standard finite element method (yielding $n_e - 1$ degrees of freedom per edge). We then enrich this construction by adding basis functions associated with the vertices, thereby incorporating the topological structure of the graph. In this way, the classical finite element basis is naturally extended to the setting of (quantum) graphs. We define the local finite element space on each edge as

$$V_h^e := \{u_h \in H^1(e) : u_h \text{ is piecewise linear with respect to the partition}\}.$$

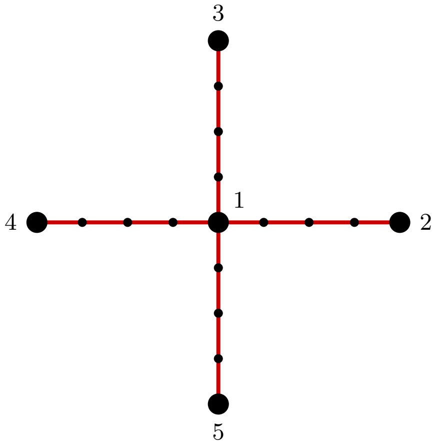
The global finite element space is then given by

$$V_h(\Gamma) := \bigoplus_{e\in\mathcal{E}} V_h(e)$$

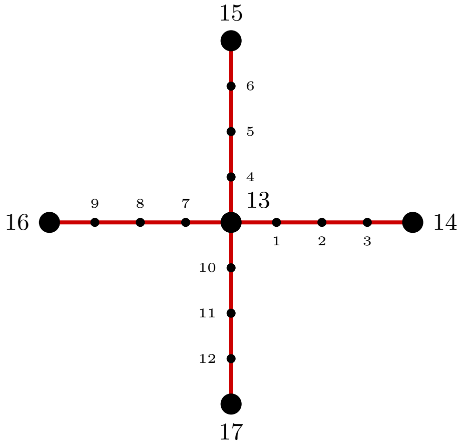
which is a finite-dimensional subspace of $H^1(\Gamma)$, in which we'll search approximate solutions to the weak formulation of differential problems. Any function $u_h \in V_h(\Gamma)$ can be written as a linear combination of the basis functions:

$$u_h(s) = \sum_{e\in\mathcal{E}} \sum_{j=1}^{n_e-1} \alpha_j^e\, \phi_j^e(s) + \sum_{v\in\mathcal{V}} \beta_v\, \phi_v(s). \tag{2}$$

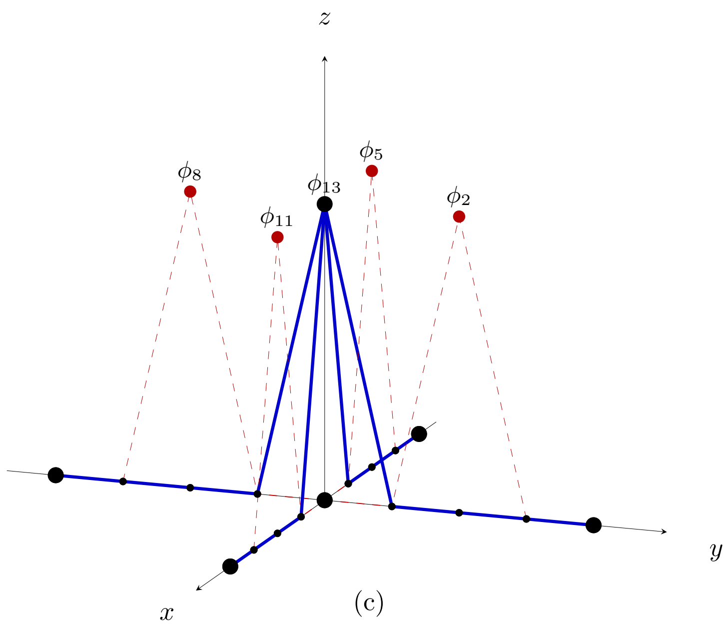
One observation that we can make is that it is convenient to order the vertices by following the original edge structure of the graph. More precisely, for each edge, the vertices introduced by the discretization (i.e., the interior nodes along that edge) are numbered consecutively. After all such nodes have been listed for every edge, the original vertices of the graph are numbered last. In this way, the vertices corresponding to the subdivided edges (the subdomain nodes) appear in contiguous blocks within linear algebra structures, while the original graph vertices (the separators) are grouped together at the end of these. This is showed in the following Figure.



(a)



(b)



(c)

Figure. (a) A star-shaped graph $\Gamma = (\mathcal{V}, \mathcal{E})$ with five vertices ($\mathcal{V} = \{v_1, v_2, v_3, v_4, v_5\}$) and four edges ($\mathcal{E} = \{\{v_1, v_2\}, \{v_1, v_3\}, \{v_1, v_4\}, \{v_1, v_5\}\}$), equipped with a uniform mesh consisting of four elements per edge. The edges incident to the central vertex v_1 are highlighted in red. (b) The same star graph with global node numbering: interior discretisation nodes are listed first, edge by edge (right: 1–3; top: 4–6; left: 7–9; bottom: 10–12, each numbered from centre outward; this orientation coincides with the directionality chosen for the construction of the incidence matrix), followed by the true graph vertices (centre: 13; right: 14; top: 15; left: 16; bottom: 17). (c) The graph Γ together with its finite-element basis. In blue, the vertex basis function ϕ_{13}; in dashed red, four representative hat basis functions, one per arm.

2 Parabolic problems on metric graphs

A comprehensive analysis on graph-based parabolic problems can be found in [4].

2.1 Problem formulation

We consider quantum graphs with parabolic Hamiltonians. Let $\Gamma = (\mathcal{V}, \mathcal{E})$ be a metric graph, $V \in L^\infty(\Gamma)$ and $f : \Gamma \to \mathbb{R}$ be given. Let $\mathbb{H}$ denote the elliptic Hamiltonian of the quantum graph $(\Gamma, \mathbb{H}, \mathcal{C}_{NK})$, with underlying Neumann-Kirchhoff conditions at each vertex in $\mathcal{V}$ and global continuity constraints. The parabolic Hamiltonian acts on functions defined on Γ as follows:

$$\mathbb{H}\, u(s,t) = \frac{\partial u(s,t)}{\partial t} + \mathcal{H} u(s,t) = \frac{\partial u(s,t)}{\partial t} - \frac{\partial^2 u(s,t)}{\partial s^2} + V(s)\, u(s,t),$$

where $\mathcal{H}$ is the elliptic Hamiltonian introduced with (Eq. 1).
Let us consider the differential problem

$$\begin{cases} \mathbb{H}\, u = f & \text{in } \Gamma \times (0,T) \\ \sum_{e \in \mathcal{E}_{v_p}} \left[\frac{\partial u_e}{\partial \mathbf{n}_e} \right]_v = 0, & \forall\, v \in \mathcal{V}, \forall\, t \in (0,T) \end{cases} \tag{3}$$

where dependence on the variables (s,t) have been omitted in the first equation.
This problem can be expressed in weak form by means of a standard procedure reported in Appendix B, yielding

$$\sum_{e \in \mathcal{E}} \int_e \frac{\partial u}{\partial t}\, w \,\mathrm{d}s + \sum_{e \in \mathcal{E}} \int_e \frac{\partial u}{\partial s} \frac{\partial w}{\partial s} \,\mathrm{d}s + \sum_{e \in \mathcal{E}} \int_e V\, u\, w \,\mathrm{d}s = \sum_{e \in \mathcal{E}} \int_e f\, w \,\mathrm{d}s, \forall w \in H^1(\Gamma). \tag{4}$$

Defining

$$a(u,w) = \sum_{e \in \mathcal{E}} \int_e \left[\frac{\partial u}{\partial s} \frac{\partial w}{\partial s} + V\, u\, w \right] \mathrm{d}s, \qquad F(w) = \sum_{e \in \mathcal{E}} \int_e f\, w \,\mathrm{d}s,$$

the weak formulation reads:

Find $u \in H^1[\Gamma \times (0,T))]$, globally continuous on Γ such that

$$\sum_{e \in \mathcal{E}} \int_e \frac{\partial u}{\partial t}\, w \,\mathrm{d}s + a(u,w) = F(w) \qquad \forall\, w \in H^1(\Gamma). \tag{5}$$

The discrete version of the problem can be obtained exploiting a standard Galerkin procedure. Let $V_h \subset H^1(\Gamma)$ be the space spanned by the basis functions $\{\phi_j^e\}_{j=1}^{n_e - 1}$ (interior node hat functions on each edge) and $\{\phi_v\}_{v \in \mathcal{V}}$ (vertex basis functions), as defined in SECTION 1.2. The discrete solution is written as in Eq.(2) and substituted into the the weak formulation (5), whose terms are therefore restricted to the subspace V_h. We first test on an interior node hat function of an arbitrary edge. The time term gives:

$$\sum_{e \in \mathcal{E}} \int_e \frac{\partial u_h}{\partial t}\, \phi_k^{e'} \,\mathrm{d}s = \sum_{j=1}^{n_{e'}-1} \dot{\alpha}_j^{e'} \underbrace{\int_{e'} \phi_j^{e'}\, \phi_k^{e'} \,\mathrm{d}s}_{P_{kj}^{e'}} + \sum_{v \in \mathcal{V}} \dot{\beta}_v \underbrace{\int_{e'} \phi_v\, \phi_k^{e'} \,\mathrm{d}s}_{R_{kv}^{e'}},$$

where only the $e = e'$ term survives due to the local support of $\phi_k^{e'}$ and $R_{kv}^{e'}$ is nonzero only for the (at most two) vertices incident to e'. The stiffness term gives:

$$a(u_h, \phi_k^{e'}) = \sum_{j=1}^{n_{e'}-1} \alpha_j^{e'} \underbrace{\int_{e'} \left[\frac{\partial \phi_j^{e'}}{\partial s} \frac{\partial \phi_k^{e'}}{\partial s} + V_{e'} \phi_j^{e'} \phi_k^{e'} \right] \mathrm{d}s}_{(H_{11})_{kj}^{e'}} +$$

$$+ \sum_{v \in \mathcal{V}} \beta_v \underbrace{\int_{e'} \left[\frac{\partial \phi_v}{\partial s} \frac{\partial \phi_k^{e'}}{\partial s} + V_{e'} \phi_v \phi_k^{e'} \right] \mathrm{d}s}_{(H_{12})_{kv}^{e'}} .$$

The forcing term gives:

$$F(\phi_k^{e'}) = \int_{e'} f_{e'}\, \phi_k^{e'}\, \mathrm{d}s.$$

We then test on an arbitrary vertex basis function. The time term gives:

$$\sum_{e\in\mathcal{E}} \int_e \frac{\partial u_h}{\partial t}\, \phi_{v'}\, \mathrm{d}s = \sum_{e\in\mathcal{E}} \sum_{j=1}^{n_e-1} \dot{\alpha}_j^e \underbrace{\int_e \phi_j^e \phi_{v'}\, \mathrm{d}s}_{R^e_{v'j}} + \sum_{v\in\mathcal{V}} \dot{\beta}_v \underbrace{\sum_{e\in\mathcal{E}} \int_e \phi_v \phi_{v'}\, \mathrm{d}s}_{Q_{v'v}},$$

where $Q_{vv'}$ is nonzero only when v and v' are the same. The stiffness term gives:

$$\begin{aligned} a(u_h, \phi_{v'}) = \sum_{e\in\mathcal{E}} \sum_{j=1}^{n_e-1} \alpha_j^e & \underbrace{\int_e \left[\frac{\partial \phi_j^e}{\partial s} \frac{\partial \phi_{v'}}{\partial s} + V_e \phi_j^e \phi_{v'} \right] \mathrm{d}s}_{(H_{21})^e_{v'j}} + \\ & + \sum_{v\in\mathcal{V}} \beta_v \underbrace{\left[\sum_{e\in\mathcal{E}} \int_e \left(\frac{\partial \phi_v}{\partial s} \frac{\partial \phi_{v'}}{\partial s} + V_e \phi_v \phi_{v'} \right) \mathrm{d}s \right]}_{(H_{22})_{v'v}}, \end{aligned}$$

while the forcing term

$$F(\phi_{v'}) = \sum_{e\in\mathcal{E}} \int_e f_e\, \phi_{v'}\, \mathrm{d}s.$$

2.2 Matrix formulation

Collecting all unknowns into $\mathbf{u} = [\mathbf{u}_{\mathcal{E}},\, \mathbf{u}_{\mathcal{V}}]^\top$, where $\mathbf{u}_{\mathcal{E}} = \big\{\{\alpha_j^e\}_{j=1}^{n_e-1}\big\}_{e\in\mathcal{E}}$ and $\mathbf{u}_{\mathcal{V}} = \{\beta_v\}_{v\in\mathcal{V}}$, the Galerkin system is

$$\mathbf{M} \begin{bmatrix} \dot{\mathbf{u}}_{\mathcal{E}}(t) \\ \dot{\mathbf{u}}_{\mathcal{V}}(t) \end{bmatrix} + \mathbf{H} \begin{bmatrix} \mathbf{u}_{\mathcal{E}}(t) \\ \mathbf{u}_{\mathcal{V}}(t) \end{bmatrix} = \begin{bmatrix} \mathbf{f}_{\mathcal{E}}(t) \\ \mathbf{f}_{\mathcal{V}}(t) \end{bmatrix}, \tag{6}$$

with

$$\mathbf{M} = \begin{bmatrix} \mathbf{P} & \mathbf{R} \\ \mathbf{R}^\top & \mathbf{Q} \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} \mathbf{H}_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{21} & \mathbf{H}_{22} \end{bmatrix}, \tag{7}$$

where the blocks' elements are defined as follows:

- $P^e_{kj} = \displaystyle\int_e \phi_j^e \phi_k^e\, \mathrm{d}s;$
- $R^e_{kv} = \displaystyle\int_e \phi_v \phi_k^e\, \mathrm{d}s;$
- $Q_{v'v} = \displaystyle\sum_{e\in\mathcal{E}} \int_e \phi_v \phi_{v'}\, \mathrm{d}s;$
- $(H_{11})^e_{kj} = \displaystyle\int_e \left[\frac{\partial \phi_j^e}{\partial s} \frac{\partial \phi_k^e}{\partial s} + V_e \phi_j^e \phi_k^e \right] \mathrm{d}s;$
- $(H_{12})^e_{vj} = \displaystyle\int_e \left[\frac{\partial \phi_j^e}{\partial s} \frac{\partial \phi_v}{\partial s} + V_e \phi_j^e \phi_v \right] \mathrm{d}s;$
- $(H_{22})_{v'v} = \displaystyle\sum_{e\in\mathcal{E}} \int_e \left[\frac{\partial \phi_v}{\partial s} \frac{\partial \phi_{v'}}{\partial s} + V_e \phi_v \phi_{v'} \right] \mathrm{d}s.$

Theorem 2.1. *The mass matrix* $\mathbf{M}$ *and the discrete Hamiltonian* $\mathbf{H}$ *have the following block structures:*

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{12}^\top & \mathbf{H}_{22} \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{G} \end{bmatrix}}_{\mathbf{L}} + \underbrace{\begin{bmatrix} \mathbf{V} & \mathbf{C} \\ \mathbf{C}^\top & \mathbf{D} \end{bmatrix}}_{\mathbf{M}_v},$$

$$\mathbf{M} = \begin{bmatrix} \mathbf{P} & \mathbf{R} \\ \mathbf{R}^\top & \mathbf{Q} \end{bmatrix},$$

where the entries of each block are defined as follows.

In particular:

1. $\mathbf{H}$ is symmetric and positive definite; $\mathbf{M}$ is symmetric positive definite.
2. $\mathbf{A}$ and $\mathbf{V}$ are block diagonal matrices, with one block per edge $e \in \mathcal{E}$, of size $(n_e - 1) \times (n_e - 1)$. Similarly, $\mathbf{P}$ is block diagonal with one block per edge.
3. $\mathbf{G}$ and $\mathbf{D}$ are diagonal matrices of size $N \times N$, being $N = |\mathcal{V}|$. Similarly, $\mathbf{Q}$ is diagonal.
4. $\mathbf{L}$, called the *stiffness matrix*, is the Laplace matrix of the extended graph. $\mathbf{M}_V$, called the *generalized mass matrix*, encodes the potential V. The full discrete Hamiltonian satisfies $\mathbf{H} = \mathbf{L} + \mathbf{M}_v$.
5. The entries of each block of $\mathbf{H}$ are:

$$A_{jk} = \int_e \frac{\partial \phi_j^e}{\partial s} \frac{\partial \phi_k^e}{\partial s}\, \mathrm{d}s = \begin{cases} 2/h_e & \text{if } j = k, \\ -1/h_e & \text{if } |j-k| = 1, \\ 0 & \text{otherwise,} \end{cases}$$

$$B_{vk} = \int_e \frac{\partial \phi_v}{\partial s} \frac{\partial \phi_k^e}{\partial s}\, \mathrm{d}s = \begin{cases} -1/h_e & \text{if } e \in \mathcal{E}_v \cap \mathcal{E}_k \neq \varnothing, \\ 0 & \text{otherwise,} \end{cases}$$

$$G_{vv} = \int_{\mathcal{W}_v} \frac{\partial \phi_v}{\partial s} \frac{\partial \phi_v}{\partial s}\, \mathrm{d}s = \sum_{e \in \mathcal{E}_v} \frac{1}{h_e},$$

$$V_{jk} = \int_e V(s)\, \phi_j^e(s)\, \phi_k^e(s)\, \mathrm{d}s, \qquad C_{vk} = \int_e V(s)\, \phi_v(s)\, \phi_k^e(s)\, \mathrm{d}s, \qquad D_{vv} = \int_{\mathcal{W}_v} V(s)\, \phi_v^2(s)\, \mathrm{d}s.$$

6. The entries of each block of $\mathbf{M}$ are:

$$P_{jk}^e = \int_e \phi_j^e\, \phi_k^e\, \mathrm{d}s = \begin{cases} 2h_e/3 & \text{if } j = k, \\ h_e/6 & \text{if } |j-k| = 1, \\ 0 & \text{otherwise,} \end{cases}$$

$$R_{kv}^e = \int_e \phi_v\, \phi_k^e\, \mathrm{d}s = \begin{cases} h_e/6 & \text{if } e \in \mathcal{E}_v \text{ and } k \text{ is the interior node adjacent to } v, \\ 0 & \text{otherwise,} \end{cases}$$

$$Q_{v'v} = \sum_{e \in \mathcal{E}} \int_e \phi_v\, \phi_{v'}\, \mathrm{d}s = \begin{cases} \displaystyle\sum_{e \in \mathcal{E}_v} \frac{h_e}{3} & \text{if } v' = v, \\ 0 & \text{otherwise.} \end{cases}$$

The diagonal structure of $\mathbf{Q}$ follows from the fact that vertex basis functions ϕ_v and $\phi_{v'}$ have disjoint support whenever $v \neq v'$, since each ϕ_v is supported only on edges incident to v.

2.3 Fully discrete system

Discretizing the semi-discrete system in time by the θ-method with uniform step $\Delta t = T/N_T$ and setting $\mathbf{u}^n \approx \mathbf{u}(t^n)$, one obtains

$$\mathbf{M}\, \frac{\mathbf{u}^n - \mathbf{u}^{n-1}}{\Delta t} + \mathbf{H}\big(\theta\, \mathbf{u}^n + (1-\theta)\, \mathbf{u}^{n-1}\big) = \theta\, \mathbf{f}^n + (1-\theta)\, \mathbf{f}^{n-1},$$

which rearranges to

$$\mathbf{M}_\theta\, \mathbf{u}^n = \big(\mathbf{M} - (1-\theta)\Delta t\, \mathbf{H}\big)\, \mathbf{u}^{n-1} + \Delta t\big(\theta\, \mathbf{f}^n + (1-\theta)\, \mathbf{f}^{n-1}\big), \tag{8}$$

where

$$\mathbf{M}_\theta \;=\; \mathbf{M} + \theta\, \Delta t\, \mathbf{H} \;=\; \begin{bmatrix} \mathbf{P} + \theta\, \Delta t\, \mathbf{H}_{11} & \mathbf{R} + \theta\, \Delta t\, \mathbf{H}_{12} \\ \mathbf{R}^\top + \theta\, \Delta t\, \mathbf{H}_{21} & \mathbf{Q} + \theta\, \Delta t\, \mathbf{H}_{22} \end{bmatrix}.$$

The choice $\theta = 1$ recovers the backward Euler method, while $\theta = \frac{1}{2}$ gives the Crank–Nicolson scheme.
Since $a(\cdot, \cdot)$ is symmetric and coercive on $H^1(\Gamma)$ and $\mathbf{M}$ is symmetric positive definite, $\mathbf{M}_\theta$ is symmetric positive definite for every $\Delta t > 0$ and $\theta \in (0, 1]$; hence the linear system admits a unique solution at each time step.

3 Numerical Benchmarks

This section presents the numerical benchmarks developed to validate the C++ implementation described in Chapter 5. The tests are organized as a progressive validation path, moving from basic elliptic checks to genuinely parabolic phenomena. Following [5], we begin with three elliptic test cases, denoted E1–E3, which serve as reference benchmarks for the parabolic extension. Test E1 verifies the correctness of the assembly procedure and the basic control flow by prescribing a constant exact solution. Test E2 assesses the enforcement of mixed Dirichlet–Neumann–Kirchhoff boundary conditions through a linear exact solution. Test E3 investigates the spatial convergence of the solver by adopting an exact solution that does not belong to the finite element space (a sinusoidal solution). We then introduce three parabolic benchmarks. Test 1 considers a radially decaying solution on a star-shaped graph, where both diffusion and the reaction term contribute to the time evolution of the solution. Test 2 assesses temporal convergence, verifying that the Backward Euler scheme achieves first-order accuracy and the Crank–Nicolson scheme achieves second-order accuracy in time. Test 3 prescribes a spatially localized initial condition and examines the qualitative propagation of the solution across the graph topology.

3.1 Elliptic Test Cases

We begin with three elliptic benchmark problems on a star-shaped metric graph Γ with five vertices and four edges. In particular, $\Gamma = (\mathcal{V}, \mathcal{E})$, with $\mathcal{V} = \{v_0 = (0,0), v_1 = (1,0), v_2 = (0,1), v_3 = (-1,0), v_4 = (0,-1)\}$ and $\mathcal{E} = \{\{v_0, v_1\}, \{v_0, v_2\}, \{v_0, v_3\}, \{v_0, v_4\}\}$. Thus, Γ consists of four unit-length edges meeting at the central vertex v_0, , like the one represented in Section 1.2.
A key observation is the following: these benchmarks, while representing elliptic cases, are used to validate a parabolic pipeline, so they will be interpreted in this setting.

3.1.1 Test E1: constant function

The first elliptic test considers a constant exact solution on the star graph:

$$u_{\text{exact}} = 5,$$

satisfying the following:

$$\frac{\partial u}{\partial t} - \nabla u = 0.$$

Boundary data and the initial condition are chosen consistently so that $u_{\text{exact}} = 5$ on every edge of Γ. This test checks the basic assembly procedure and control flow of the code for a trivial case. At final time $T = 1$, the numerical solution remains constant on the whole graph, with value $u = 5$ up to roundoff errors. In this simple case, the numerical algorithms are expected to return the exact solution. This is indeed the case, as shown in the following Figure. A mesh with 100 elements per edge is used to discretize the graph. The graph is discretized again using 100 elements per edge. The final time of observation is $T = 1$, the time step is $\Delta t = 0.05$ and the Backward Euler scheme is used.

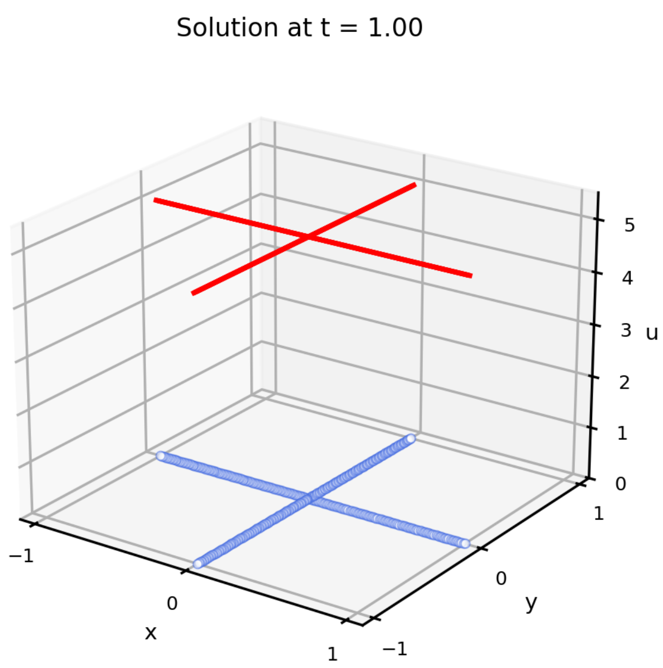


Figure 1: Test E1. Constant solution at final time $T = 1$, with the computational star graph drawn in blue at height zero and the numerical solution lifted in red. The flat profile confirms that $u = 5$ is preserved on every edge of the graph.

3.1.2 Test E2: linear function

Let us consider the parabolic problem

$$\begin{cases} \frac{\partial u}{\partial t} - \Delta u = 0 & \text{in } \Gamma \times (0, T), \\ \displaystyle\sum_{e \in \mathcal{E}_{v_0}} \frac{\partial u_e}{\partial n_e}(v_0) = 0, & \\ u(v_1, t) = 1, & t \in (0, T), \\ u(v_2, t) = 1, & t \in (0, T), \\ u(v_3, t) = -1, & t \in (0, T), \\ u(v_4, t) = -1, & t \in (0, T), \\ u(x, y, 0) = x + y, & \text{on } \Gamma. \end{cases}$$

The exact solution is stationary and is given by

$$u_{\text{ex}}(x, y) = x + y.$$

Since the exact solution is linear on each edge, it belongs to the finite element space of continuous piecewise linear functions. Therefore, the numerical method is expected to reproduce it up to roundoff errors. The graph is discretized again using 100 elements per edge. The final time is still $T = 1$, the time step is $\Delta t = 0.05$ and the Backward Euler scheme is used. This benchmark proves that our code can handle various boundary conditions.

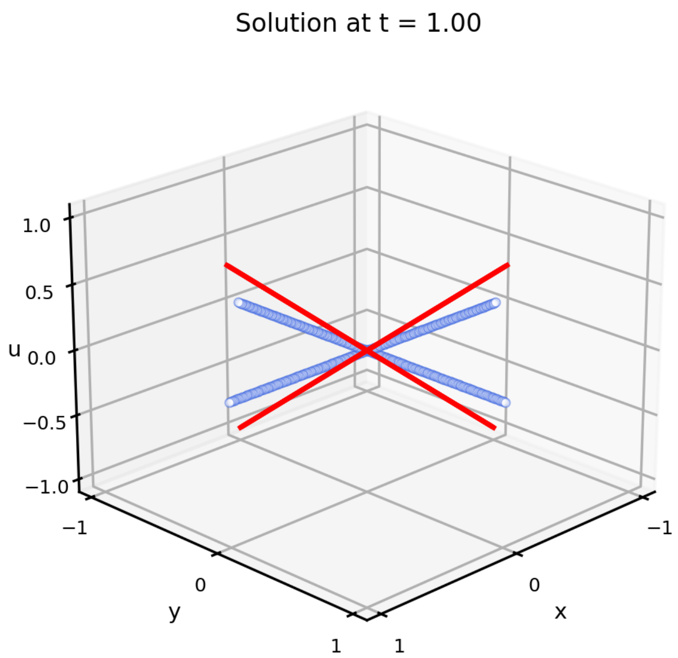


Figure 2: Test E2. Linear solution at final time $T = 1$. The stationary profile $u = x + y$ is preserved on the whole graph: the solution is linear with unit slope on all four branches, running from -1 to 1 along both the horizontal and the vertical path and continuous at the central vertex.

3.1.3 Test E3: function of a single variable

The third elliptic test considers a non-polynomial function depending only on the horizontal coordinate:

$$u_{\text{ex}}(x, y) = \sin(2\pi x).$$

This solution is not exactly represented by the piecewise linear finite element space. Therefore, unlike the previous two tests, the numerical error is nonzero and should decrease as the mesh is refined. This test is useful to observe the spatial convergence properties of the method.

On the star-shaped graph Γ, we solve

$$\begin{cases} \partial_t u - \Delta u = 4\pi^2 \sin(2\pi x) & \text{in } \Gamma \times (0, T), \\ u(v, t) = 0 & \forall v \in \mathcal{V},\ t \in (0, T), \\ u(x, y, 0) = \sin(2\pi x) & \text{on } \Gamma. \end{cases}$$

Unlike the constant and linear benchmarks, the exact solution is not contained in the finite element space of continuous piecewise linear functions. Therefore the numerical error is not expected to vanish, but it should decrease as the mesh is refined. The final time is $T = 1$, the time step is $\Delta t = 0.005$ and the Backward Euler method is used.

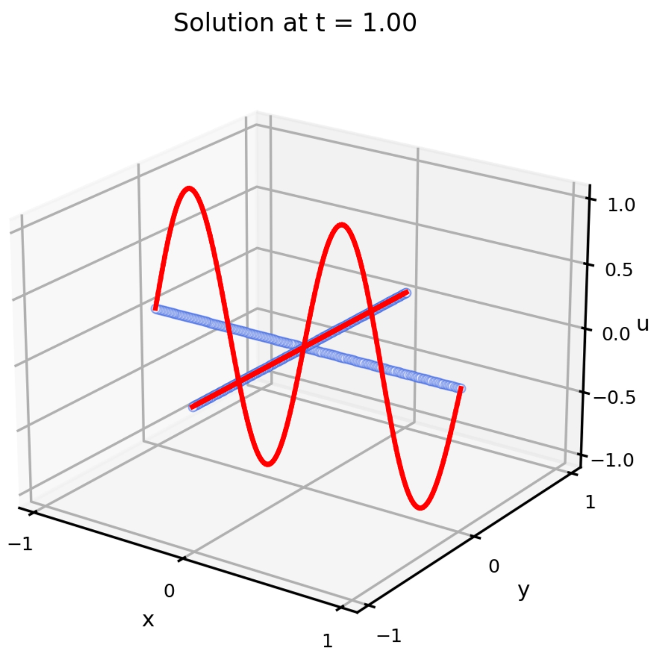


Figure 3: Test E3. Numerical solution associated with $u_{\mathrm{ex}}(x, y, t) = \sin(2\pi x)$, lifted in red above the star graph. The two full periods develop along the horizontal branch, the solution is zero on the vertical branches, where $x = 0$, and the discrete solution is correctly globally continuous at the central vertex, where the sine vanishes.

To study the spatial convergence, the problem is solved using 10, 20, 40, and 80 elements per edge. The time step is chosen as $\Delta t = 0.025h^2$, so that the temporal error is negligible with respect to the spatial error. Since continuous piecewise linear finite elements are used, the interpolation error for a sufficiently regular exact solution satisfies the classical estimate: $\|u - u_h\|_{L^2(\Gamma)} \leq Ch^2 \|u\|_{H^2(\Gamma)}$. Therefore, when the temporal error is made negligible by choosing $\Delta t = 0.025h^2$, the dominant contribution to the error is the spatial finite element error. Hence, second-order convergence in the L^2-norm is expected. The numerical results confirm this behaviour, since the observed convergence rates are approximately equal to 2.

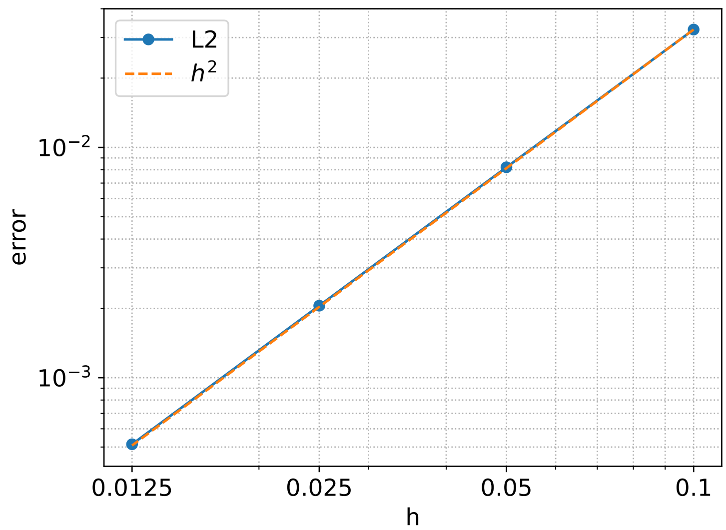


Figure 4: Test E3. Spatial convergence for the parabolic sine benchmark. The results show second-order convergence in the L^2-norm.

3.2 Parabolic Test Cases

We now present the parabolic benchmarks implemented in this project.

3.2.1 Test 1:

This benchmark introduces a genuinely time-dependent exact solution. We solve

$$\partial_t u - \Delta u + u = 0 \qquad \text{on } \Gamma \times (0, T).$$

The exact solution on each edge is

$$u_{\mathrm{ex}}(s,t) = \exp\left[-\left(\frac{\pi^2}{4} + 1\right) t\right] \cos\left(\frac{\pi s}{2}\right).$$

Homogeneous Dirichlet conditions are imposed at the four external vertices, while the central vertex satisfies the natural Kirchhoff condition. This benchmark validates the time-dependent part of the solver.

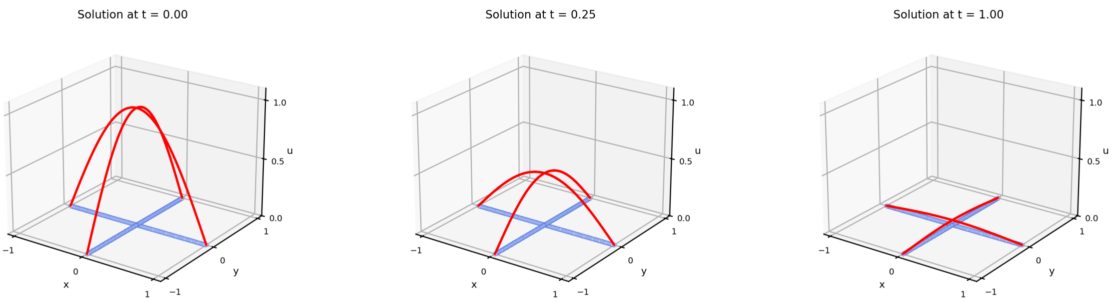


Figure 5: This test considers a radially decaying solution where both the diffusion and reaction term contribute to the temporal dynamics of the solution: the initial bump $\cos(\pi s/2)$, equal to one at the central vertex and pinned at zero at the four external vertices, decays uniformly on the star. The Crank–Nicolson scheme ($\theta = 1/2$) is used, with $\Delta t = 0.05$ and a mesh size $h = 0.01$.

For the temporal convergence test, a fine spatial mesh with 320 elements per edge is used. Since each edge has unit length, the mesh size is $h = \frac{1}{320} = 0.003125$. This choice makes the spatial error negligible with respect to the temporal error. Indeed, for piecewise linear finite elements the L^2-error behaves as $O(h^2)$, and $h^2 \simeq 9.77 \cdot 10^{-6}$. On the other hand, the smallest time step used in the test is $\Delta t = 0.025$, for which $\Delta t^2 = 6.25 \cdot 10^{-4}$. Therefore, even for the second-order Crank–Nicolson method, the temporal error is expected to dominate the spatial one.

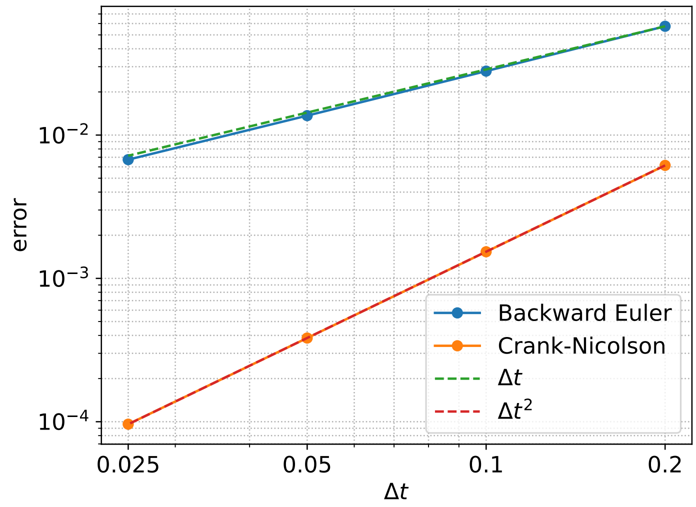


Figure 6: Test 4. Temporal convergence of Backward Euler and Crank–Nicolson. The results show first-order convergence for Backward Euler and second-order convergence for Crank–Nicolson.

3.3 Elliptic eigenvalue problems

The following numerical experiments consider the eigenvalue problem for the Laplacian operator on metric graphs, endowed with Neumann–Kirchhoff conditions at the vertices. Three different graph topologies are analyzed in order to compare our numerical results with the reference ones reported in [6]. The comparison is understood up to the sign and normalization of the eigenfunctions. Moreover, in the case of multiple eigenvalues, the computed eigenfunctions may differ by linear combinations within the same eigenspace.

3.3.1 Test 1: four-pointed star graph

The goal of this benchmark, as for the following ones, is to verify that the code correctly solves generalized eigenvalue problems on metric graphs. This also represents the first step of the parabolic pipeline, since an eigenfunction of the graph Laplacian provides an exact modal solution of the heat equation. The first topology considered is the four-pointed star graph. We compute the eigenpairs (λ, u) satisfying

$$-\Delta u = \lambda u \qquad \text{on } \Gamma,$$

with homogeneous Neumann conditions at the four external vertices and the Neumann–Kirchhoff condition at the central vertex. Each edge has length 1 and is discretized using 100 finite elements. Fig. 7 shows the first six eigenmodes. The blue lines represent the metric graph, while the red curves represent the corresponding eigenfunctions lifted in the vertical direction. The eigenfunctions are rescaled only for visualization purposes; the eigenvalues are those computed by the finite element generalized eigenvalue problem. The results are in good agreement with the reference results reported on page 31 of [6].

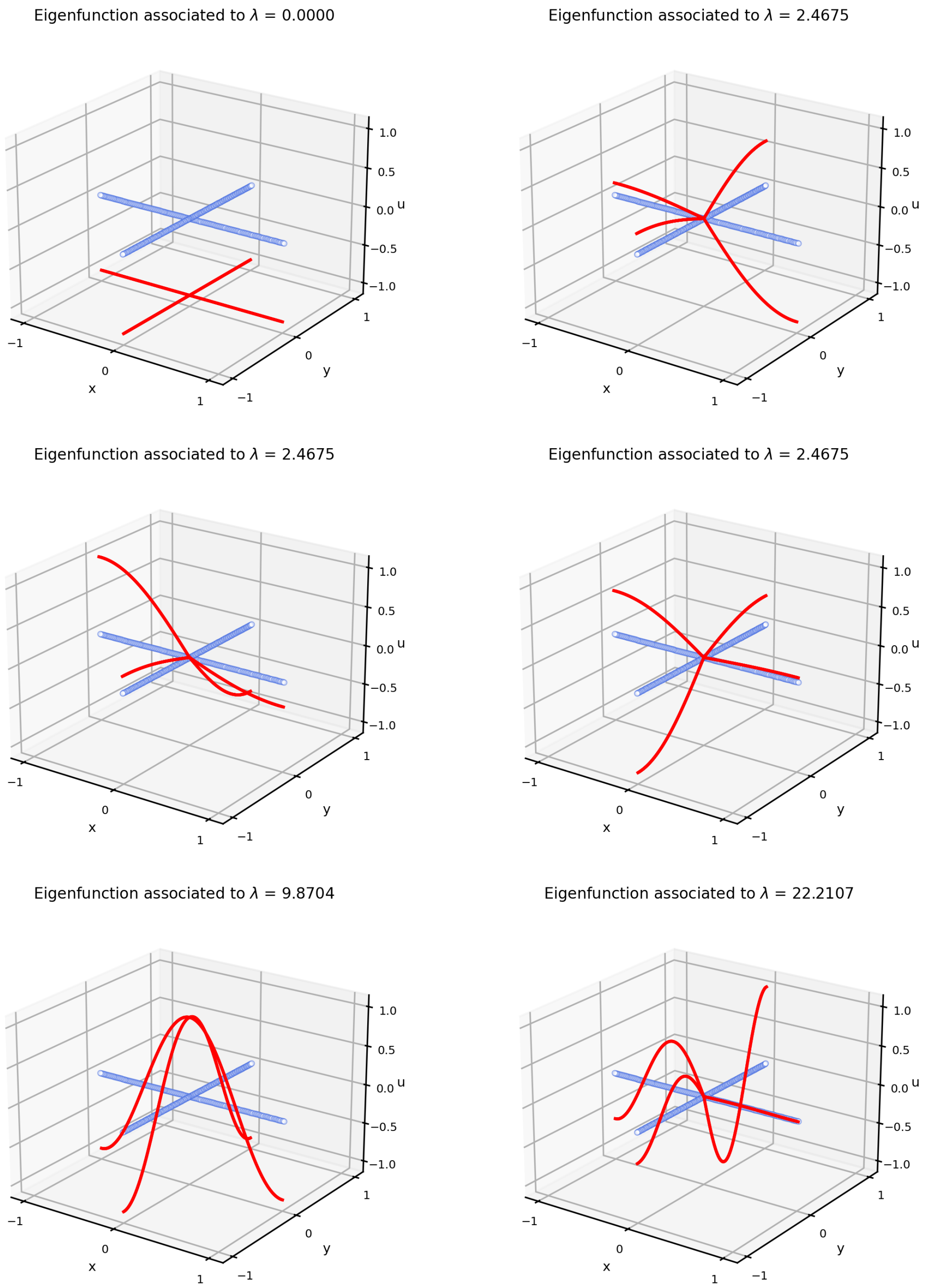


Figure 7: First six eigenmodes of the Laplacian on the four-pointed star graph. The blue graph represents the computational domain, while the red curves show the corresponding eigenfunctions. The vertical scaling is used only for visualization.

The eigenvectors computed by the finite-element solver are normalized with respect to the mass matrix, namely $\mathbf{u}^T M \mathbf{u} = 1$, which corresponds to the discrete $L^2(\Gamma)$-normalization. In the figures, however, each eigenfunction is additionally rescaled by its maximum absolute nodal value only for visualization purposes.

3.3.2 Test 2: graphene-like graph

The second benchmark considers a graphene-like metric graph. This topology is more complex than the four-pointed star graph, since it contains several internal junctions and cycles. The aim is to test our solver on a graph with a richer connectivity structure, while keeping the same differential operator and vertex conditions introduced in the previous test. We compute the eigenpairs (λ, u)

satisfying

$$-\Delta u = \lambda u \qquad \text{on } \Gamma,$$

with homogeneous Neumann conditions at the external vertices and Neumann–Kirchhoff conditions at the internal vertices. The graph has 12 vertices and 13 edges. Each edge has length 1 and is discretized using 100 elements 100 per edge.

Fig. 8 shows the first six eigenmodes, while Figure 9 shows the following four. The blue lines still represent the metric graph, whereas the red curves represent the computed eigenfunctions lifted in the vertical direction. As in the previous test, the vertical scaling is used only for visualization purposes. The two related images show good agreement with [6].

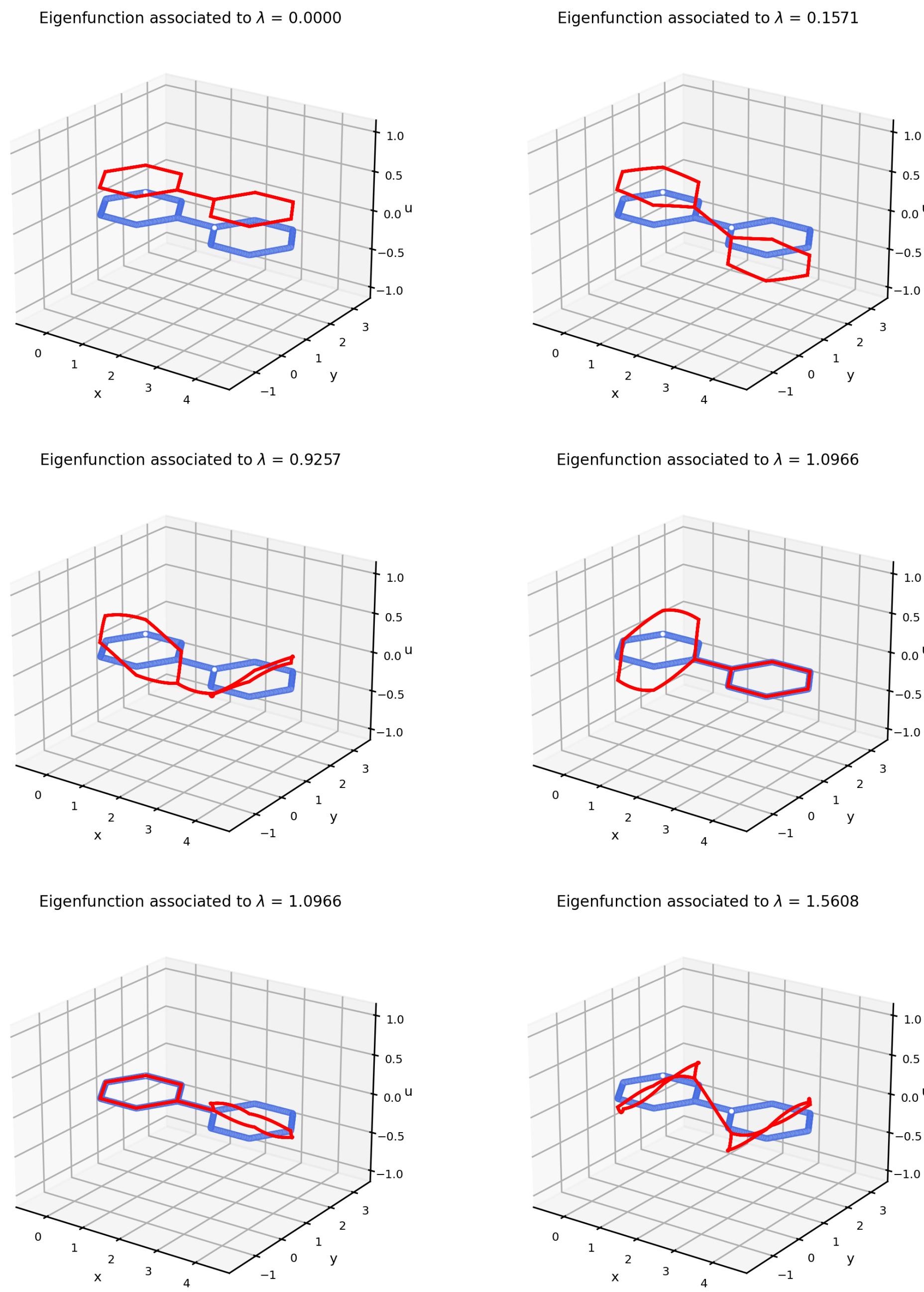


Figure 8

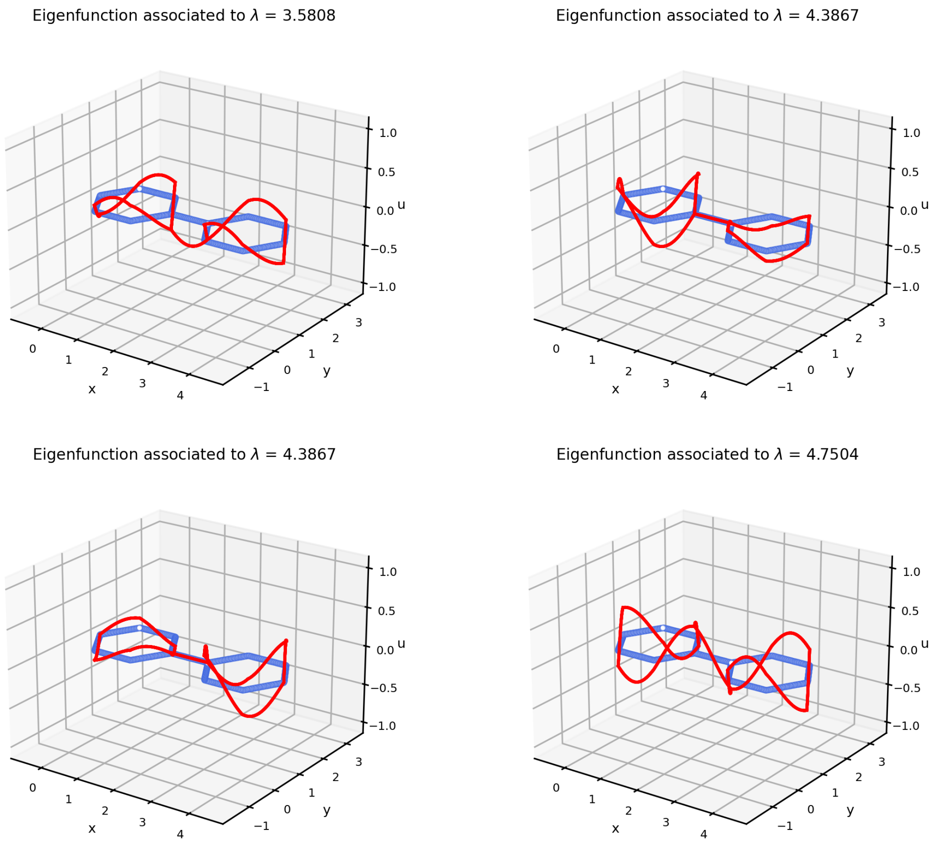


Figure 9: Fig. 9 and 10 show the first ten eigencouples of the Laplacian operator on a graphene-like graph. In blue, the domain of the problem. Results are scaled.

3.3.3 Test 3: binary tree graph

The third benchmark considers a binary tree metric graph. This topology is particularly relevant because it contains several branching points and many terminal vertices, while remaining simpler than a general network. It is therefore a useful intermediate test between elementary graph topologies and more complex network-like domains.
We compute again the eigenpairs (λ, u) satisfying

$$-\Delta_\Gamma u = \lambda u \qquad \text{on } \Gamma,$$

with homogeneous Neumann conditions at the leaves and Neumann–Kirchhoff conditions at the internal ones. The graph has 16 vertices and 15 edges. Each edge has length 1 and is discretized using 100 finite elements as in the previous cases.
Fig. 10 and Fig. 11 show the first sixteen eigenmodes. As before, the blue lines represent the metric graph and the red curves represent the eigenfunctions lifted in the vertical direction. The vertical scaling is used only for visualization. This experiment shows a very good agreement with the reference spectral behaviour reported in [6].

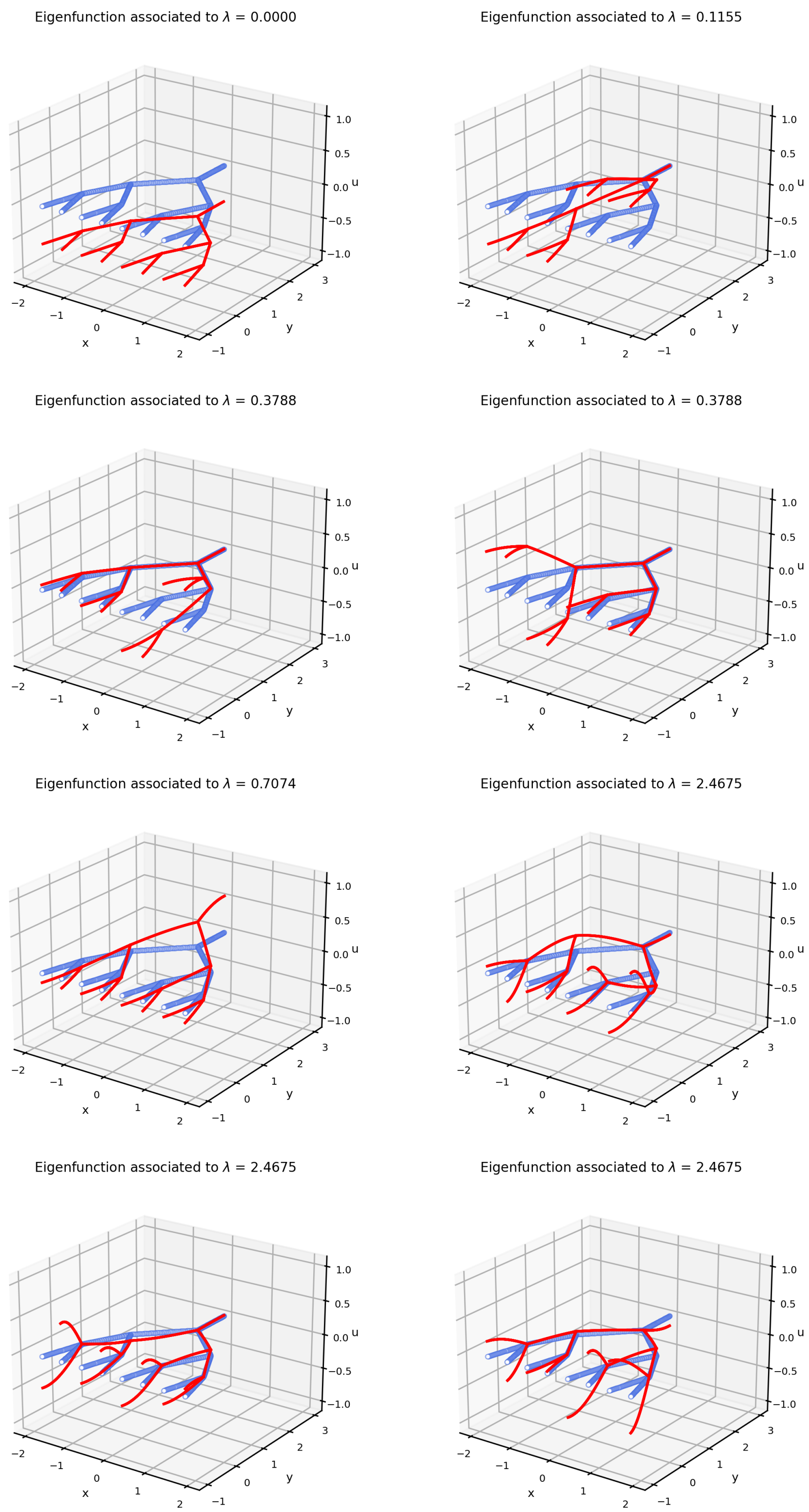


Figure 10: First eight eigenmodes of the Laplacian on the binary tree graph.

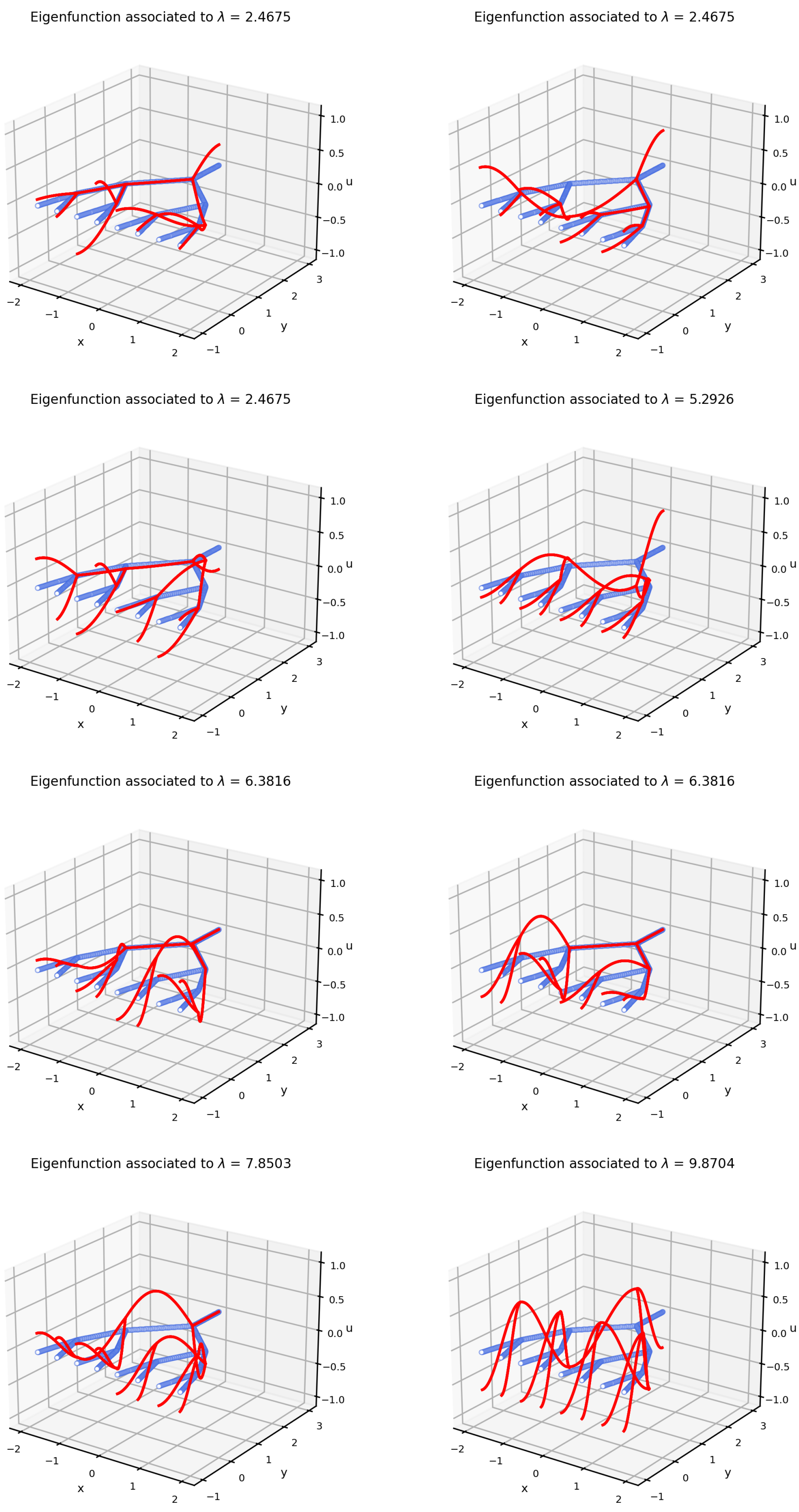


Figure 11: Eigenmodes 8–15 of the Laplacian on the binary tree graph.

Remark. In all the experiments, eigenfunctions associated with larger eigenvalues display a progressively more oscillatory behaviour along the graphs. This is consistent with the spectral properties of the Laplacian operator: the operator is unbounded, meaning that its spectrum is not bounded from above and $\lambda_k \to +\infty$ as $k \to \infty$; therefore, higher eigenvalues correspond to higher-frequency modes on the metric graph.

3.4 Parabolic eigenvalue problems

The eigenvalue computations of the previous section can be used to construct exact solutions for the heat equation on metric graphs. Let u_k be an eigenmode of the graph Laplacian associated with the eigenvalue λ_k, namely

$$-\Delta u_k = \lambda_k u_k \qquad \text{on } \Gamma,$$

with Neumann–Kirchhoff vertex conditions. Then the heat equation

$$\partial_t u - \mu \Delta u = 0 \qquad \text{on } \Gamma \times (0, T]$$

admits the exact modal solution

$$u(x,t) = e^{-\mu \lambda_k t} u_k(x).$$

This provides a useful validation test for the parabolic solver, since both the spatial structure and the temporal decay are known explicitly. The initial datum is a normalized eigenmode, so that

$$\|u_k\|_{L^2(\Gamma)} = 1.$$

Consequently, the exact evolution of the L^2-norm is

$$\|u(t)\|_{L^2(\Gamma)} = e^{-\mu \lambda_k t}.$$

In the following tests we take $\mu = 1.0$, $T = 1$, $\Delta t = 0.01$, and use the Crank–Nicolson method.

3.4.1 Test 4: heat equation on the graphene-like graph

For the graphene-like graph we select the sixth eigenmode, associated with $\lambda_6 \simeq 3.5808$. Fig. 12 shows the numerical solution at selected time instants. The shape of the selected eigenmode is preserved, while its amplitude decays in time. Fig. 13 compares the numerical L^2-norm with the exact decay law. Since the initial eigenmode is mass-normalized, the curve starts from 1. At $T = 1.0$, the numerical norm is approximately $2.7842 \cdot 10^{-2}$, in excellent agreement with the exact value $2.7853 \cdot 10^{-2}$.

3.4.2 Test 5: heat equation on the binary tree graph

For the binary tree graph we select the fifth eigenmode, associated with $\lambda_5 \simeq 2.4675$. Fig. 14 shows the time evolution of the numerical solution. Also in this case, the eigenmode profile is preserved and only its amplitude decays. Fig. 15 shows the agreement between the numerical L^2-norm and the exact exponential decay. At $T = 1.0$, the numerical norm is approximately $8.4790 \cdot 10^{-2}$, while the exact value is $8.4801 \cdot 10^{-2}$.

3.4.3 Laplace matrix spectral comparison

It can be interesting to compare the spectrum of the combinatorial Laplacian of the original graph Γ with the spectrum of the Laplace matrix associated with the corresponding extended graph $\mathcal{G}$. Given a graph with N vertices, the combinatorial Laplacian, introduced in Definition 1.6 (and described in App. A), is

$$L = D - A,$$

where D is the degree matrix and A is the adjacency matrix. In our implementation, this matrix is assembled through a custom C++ routine from the graph connectivity. Since L is real and symmetric, its eigenvalues are then computed using Eigen's `SelfAdjointEigenSolver()`.
For the extended graph, the finite element discretization of the Laplace eigenvalue problem leads to the generalized algebraic problem

$$H\mathbf{u} = \lambda M \mathbf{u},$$

where H is the stiffness matrix and M is the mass matrix. This problem is solved using Eigen's `GeneralizedSelfAdjointEigenSolver()`. This choice is motivated by the fact that H and M are symmetric matrices and M is positive definite.
The comparison is performed on three graph topologies: the four-pointed star graph, the graphene-like graph and a family of tree graphs. The extended graph is constructed using 100 finite elements per edge. This choice makes the spectral comparison consistent with the eigenfunction plots

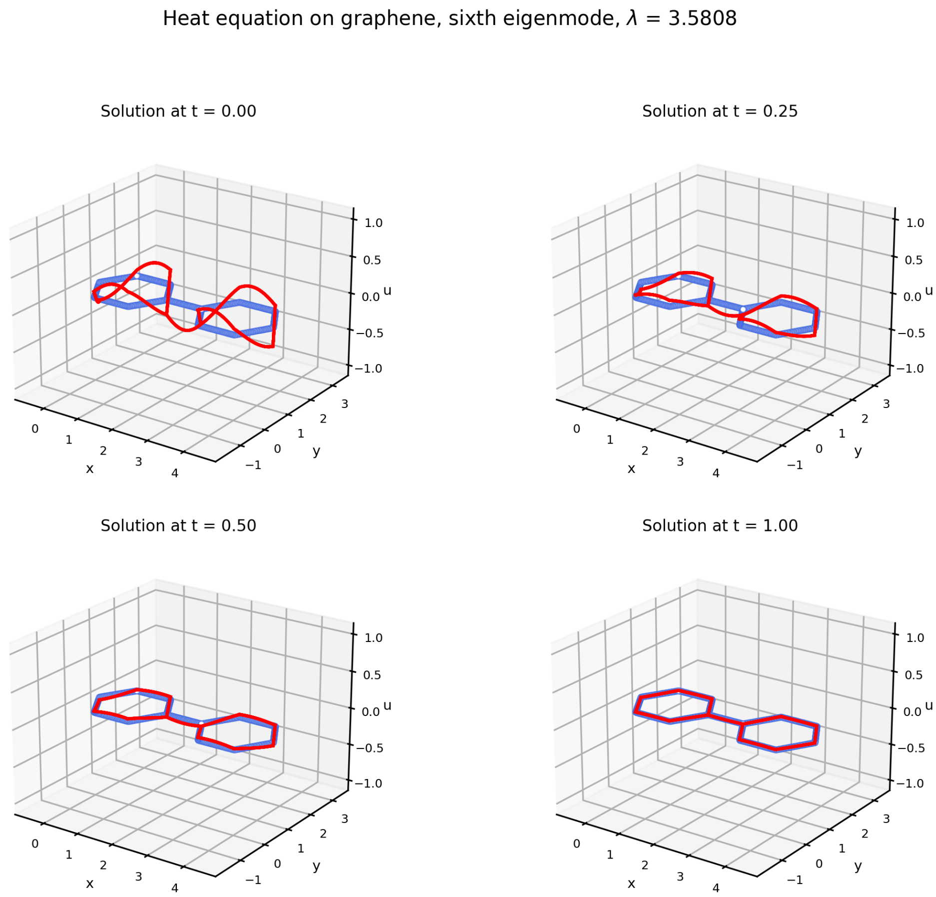


Figure 12: Heat equation on the graphene-like graph with initial datum given by the sixth eigenmode, associated with $\lambda_6 \simeq 3.5808$.

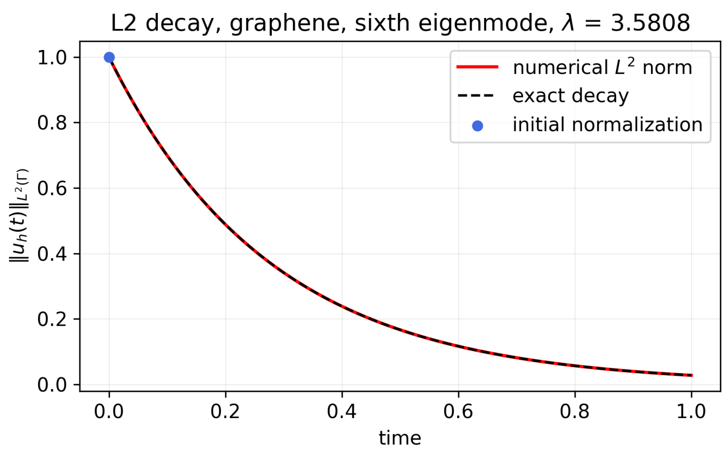


Figure 13: Decay of the numerical L^2-norm compared with the exact law for the graphene-like graph.

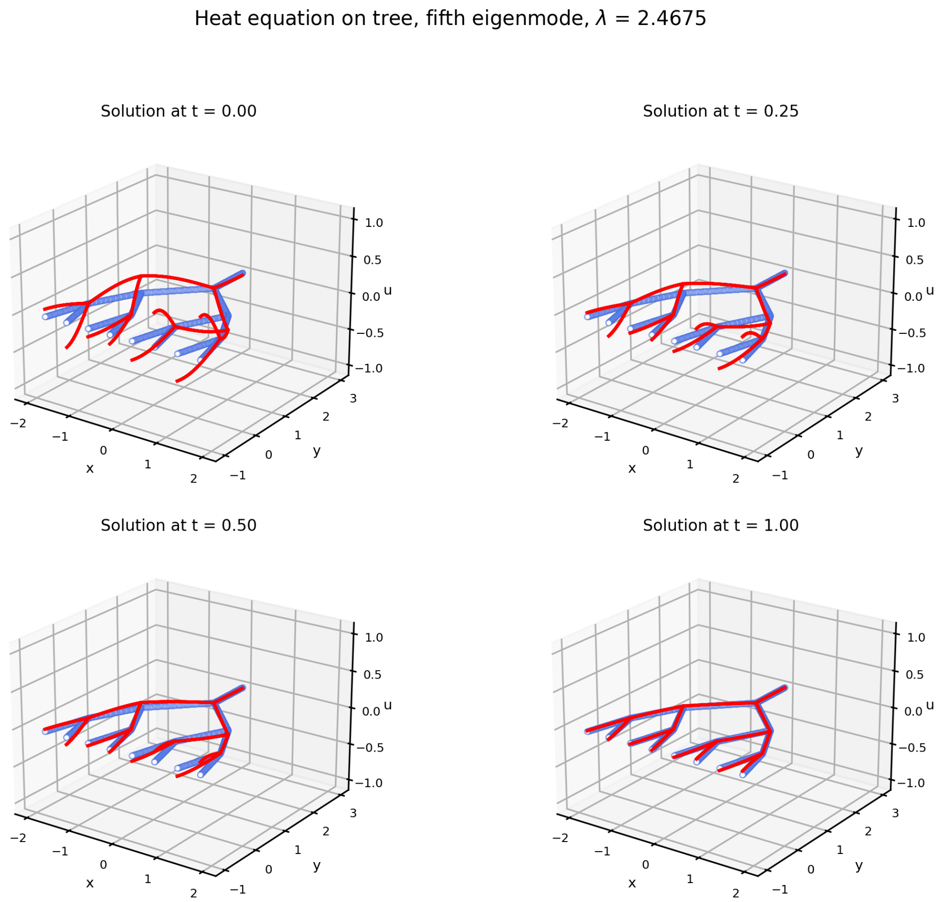


Figure 14: Heat equation on the binary tree graph with initial datum given by the fifth eigenmode, associated with $\lambda_5 \simeq 2.4675$.

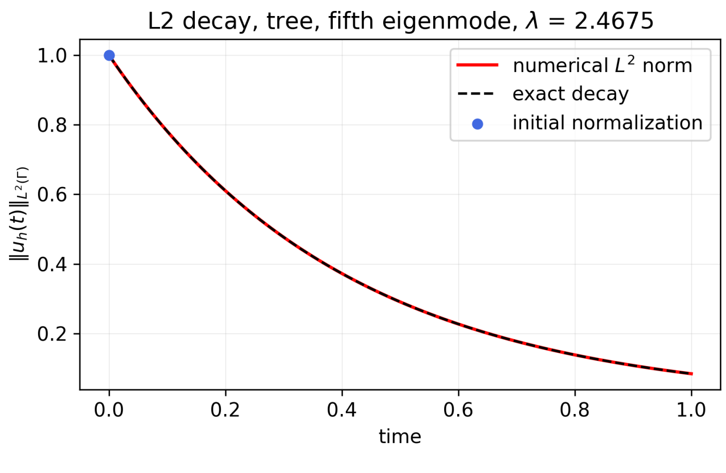


Figure 15: Decay of the numerical L^2-norm compared with the exact law for the binary tree graph.

presented in the previous benchmarks.
Fig. 16 reports the comparison between the eigenvalues of the combinatorial Laplace matrix and those of the extended Laplace matrix. The corresponding numerical values are given in Tables 1, 2 and 3.

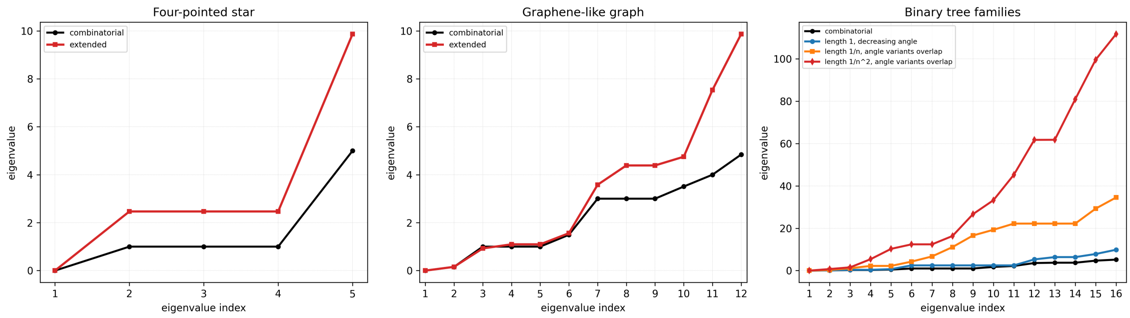


Figure 16: Comparison between the spectrum of the combinatorial Laplace matrix of the original graph and the spectrum of the extended Laplace matrix. Eigenvalues are plotted against their index. The presence of symmetries in the graph topology causes the existence of multiple eigenvalues.

Combinatorial Laplace matrix	Extended Laplace matrix
0	0
1.0000	2.4675
1.0000	2.4675
1.0000	2.4675
5.0000	9.8704

Table 1: Spectral comparison for the four-pointed star graph.

Combinatorial Laplace matrix	Extended Laplace matrix
0	0
0.1578	0.1571
1.0000	0.9257
1.0000	1.0966
1.0000	1.0966
1.4931	1.5608
3.0000	3.5808
3.0000	4.3867
3.0000	4.3867
3.5069	4.7504
4.0000	7.5371
4.8422	9.8704

Table 2: Spectral comparison for the graphene-like graph.

For the tree graph, several geometric configurations are considered. The first case has constant edge length and decreasing bifurcation angle. The other cases have either linearly or quadratically (with the bifurcation level) decreasing edge lengths. More precisely, if n denotes the bifurcation level, the edge lengths are chosen as

$$\ell_n = 1, \qquad \ell_n = \frac{1}{n}, \qquad \ell_n = \frac{1}{n^2}.$$

The numerical experiments show that, for fixed and decreasing edge-length laws, changing the bifurcation angle does not affect the spectrum of the extended Laplace matrix. This is expected, since the metric graph Laplacian depends on the graph topology and on the edge lengths, while the planar drawing angles only affect the visualization of the graph.
The results show that the spectra of the combinatorial and extended Laplace matrices are generally different in value. Nevertheless, the comparison is useful to observe how the graph topology influences

Comb.	$\ell_n = 1$	$\alpha = \pi/4,\ \ell_n = 1/n$	$\alpha = \pi/4,\ \ell_n = 1/n^2$
0	0	0	0
0.0968	0.1155	0.3303	0.7426
0.2679	0.3788	1.1186	1.5485
0.2679	0.3788	2.2205	5.4138
0.4965	0.7074	2.2205	10.2825
1.0000	2.4675	4.2573	12.4108
1.0000	2.4675	6.7012	12.4108
1.0000	2.4675	11.1094	16.3523
1.0000	2.4675	16.5789	26.6628
1.7356	2.4675	19.2819	33.2463
2.1939	2.4675	22.2071	45.2803
3.5767	5.2926	22.2071	61.7503
3.7321	6.3816	22.2071	61.7907
3.7321	6.3816	22.2071	80.9487
4.7093	7.8503	29.2923	99.6367
5.1912	9.8704	34.5652	111.7665

Table 3: Spectral comparison for the tree graph. The omitted angle variants give the same extended spectra for fixed edge-length laws.

the low-frequency part of the spectrum. In the tree cases, the dominant effect is produced by the edge lengths: decreasing edge lengths lead to larger eigenvalues, while changing the bifurcation angles does not modify the spectrum.

An additional observation concerns the effect of decreasing edge lengths. When the tree contains shorter and shorter edges, the eigenvalues of the extended Laplace matrix increase significantly. This is consistent with the metric nature of the operator: small edges allow eigenfunctions with rapid spatial oscillations and these high-frequency modes correspond to larger eigenvalues. Therefore, the growth of the spectrum reflects the increasing presence of small geometric scales in the graph.

4 Spreading of neurodegenerative diseases within the brain's connectome

We have validated our finite element library for the solution of graph-based differential problems on a sequence of elliptic and parabolic benchmarks. We now consider a more application-oriented test case. In particular, we study the Fisher–Kolmogorov equation on a brain connectome's topology. This problem provides a natural extension of the previous linear parabolic models, as it combines diffusion along the graph with a nonlinear reaction term. At the same time, it provides a meaningful test for assessing the scalability of our framework through graph reordering and parallel programming techniques, with the particularly interesting perspective of modeling the propagation of infectious misfolded proteins through the brain, a process that is responsible for neurodegenerative diseases. The goal of this section is therefore not to provide a detailed biological study, but to investigate how the developed solver can be extended from controlled benchmark problems to a realistic network-based diffusion-reaction model.

4.1 One-dimensional sensitivity analysis

Prion diseases are characterized by a chain reaction in which misfolded proteins force native ones into a similar pathogenic structure. A physics-based reaction-diffusion model can explain the spreading and growth of these proteins in neurodegenerative diseases: the Fisher-Kolmogorov equation combined with an anisotropic diffusion model [7, 8, 9]. Following the work described in [7], we demonstrate the role of the individual model parameters through a one-dimensional sensitivity test, that we exploit also to validate the nonlinear solver before it touches a connectome. We solve

$$\frac{\partial c}{\partial t} - d\,\frac{\partial^2 c}{\partial x^2} - \alpha\, c\,(1-c) = 0, \qquad (x,t) \in (-1,1) \times (0,20], \tag{9}$$

with homogeneous Neumann boundary conditions,

$$\frac{\partial c}{\partial x}(-1,t) = \frac{\partial c}{\partial x}(1,t) = 0.$$

The initial concentration is zero at every mesh node except at the central one, where $c(0,0) = 0.1$. The interval is discretized using 200 linear finite elements. Backward Euler is employed with $\Delta t = 0.1$ and the nonlinear problem arising at each time step is solved by Newton's method; this is the fully implicit scheme. The sensitivity analysis combines the diffusion coefficients $d \in \{1.0, 2.0, 4.0\} \cdot 10^{-4}$ with the reaction coefficients $\alpha \in \{1.0, 2.0, 4.0\}$. These are the choices made in [7], including the number of elements and the number of time steps. Figure 17 presents the results using an activation-time representation: for every pair (x, c), the color denotes the first time at which the concentration at position x reaches a prescribed level of c. The black curves are therefore contours of equal activation time. Increasing α accelerates the local conversion of healthy proteins, whereas increasing d enhances their spatial propagation; consequently, both parameters increase the propagation speed, although through different mechanisms. The results are in agreement with the ones of [7].

From this sensitivity test, we can derive the main properties of prion-like diseases:

- the progression of the disease is inevitable if there is an initial infectious seeding;
- the disease progression is characterized by a local increase at the initial seeding region and a subsequent global spread across the domain;
- the critical amount of misfolded proteins (quantity that defines the end of the incubation period) is reached at the seeding site as soon as the initial concentration and growth rate α are higher.

The reference reports a preliminary study in which the time step was varied from $\Delta t/4$ to $4\Delta t$, using $\Delta t = 0.1$, with $\alpha = 2.0$ and $d = 2 \cdot 10^{-4}$. The authors concluded that $\Delta t = 0.1$ was sufficiently small to obtain a converged solution, whereas larger time steps produced a spurious increase in spreading. That study is described in words only: neither the data nor a figure is given, so the used criterion behind the statement cannot be reconstructed. What we can do is to repeat the experiment in

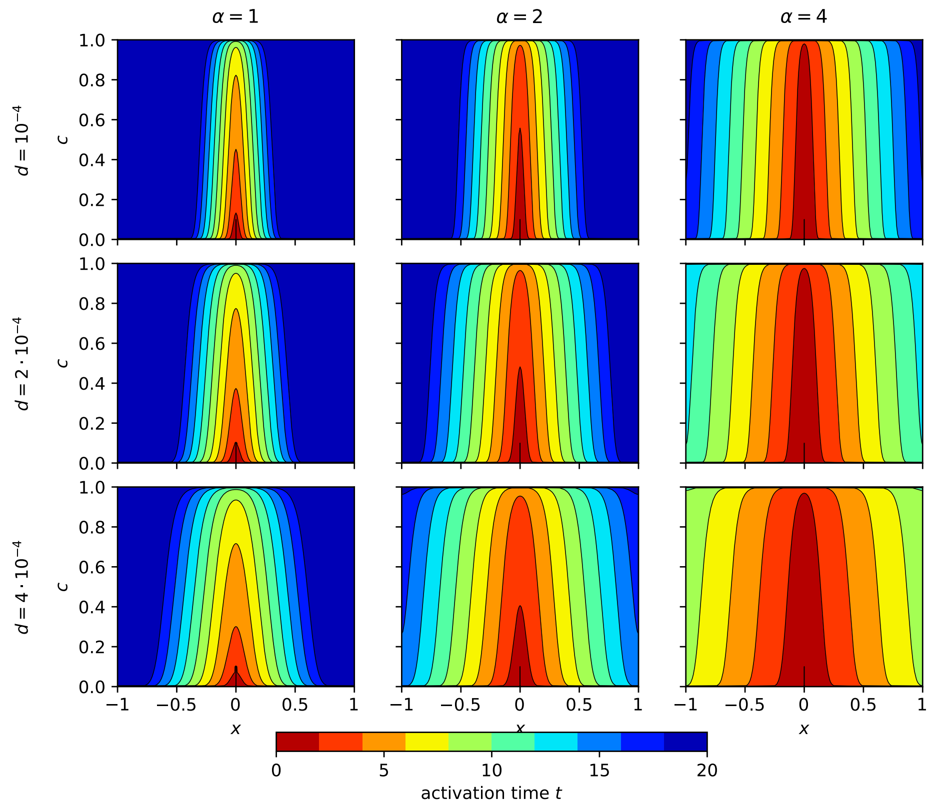


Figure 17: Activation-time maps for the one dimensional Fisher–Kolmogorov problem. The reaction coefficient increases from left to right, while the diffusion coefficient increases from top to bottom. For each pair (x, c), the color represents the first time at which the numerical concentration at x reaches a particular level c; black curves denote equal activation times. All panels use the same time scale $t \in [0, 20]$.

order to observe the spurious solutions obtained for larger time steps. We therefore resort to the coordinate x_f of the right front of the solutions as a diagnostic reference, defined as the point where the profile of the curve crosses $c = 0.5$. All the runs are compared at the common time $T = 19.2$, the largest time that is a whole multiple of every step tested: $\Delta t = \{0.025, 0.05, 0.1, 0.2, 0.3, 0.4\}$. Figure 18 shows the outcome of the experiment. The qualitative observation of the reference is confirmed. Increasing the time step moves the front outward and at a certain point, the front disappears altogether: for $\Delta t \geq 0.2$ the interval is essentially full, so no $c = 0.5$ crossing exists at all. Large time steps do produce a spurious increase in numerical propagation, exactly as described. Quantitatively, the reading is truly delicate: the three time steps that admit a front give

$$x_f = \{0.6889, \quad 0.7268, \quad 0.8145\} \qquad \text{for} \qquad \Delta t = \{0.025,\ 0.05,\ 0.1\},$$

so the solution at $\Delta t = 0.1$ is not yet insensitive to further temporal refinement. This does not directly contradict the reference, since the criterion used there to assess convergence is not reported. Our numerical result only shows that, when the front position is used as a diagnostic, the solution still changes appreciably under further temporal refinement.

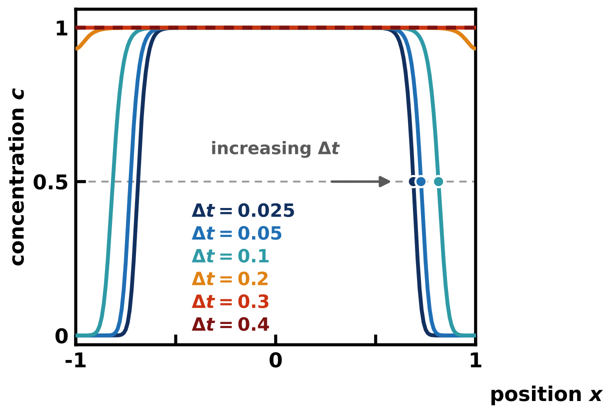


Figure 18: Time-step sensitivity plot at $\alpha = 2.0$, $d = 2 \cdot 10^{-4}$. All the runs are compared at the common time $T = 19.2$. Here, we show the six final profiles, with the $c = 0.5$ level marked by the dashed line and the resulting front positions by the circles. Only three runs cross that level. For $\Delta t \geq 0.2$ the interval is full and no crossing exists. The profile at $\Delta t = 0.4$ is dashed because it basically coincides with the one at $\Delta t = 0.3$. On the plateau $|x| \leq 0.5$ all six runs agree, so the disagreement is confined to the fronts.

4.2 The brain connectome as a metric graph

Having verified the nonlinear solver on a controlled one-dimensional problem, we now move to the network on which the disease is assumed to spread, reflecting the prion-like hypothesis that misfolded proteins propagate through the brain's axonal fiber tracts. Following [9], we represent the brain as a weighted undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ whose vertices are anatomical regions and whose edges are the axonal bundles connecting them. The graph is extracted from the public Budapest Reference Connectome v3.0 [10], which aggregates the tractography of 418 healthy subjects of the Human Connectome Project. The downloaded fine-resolution graph contains 1015 parcels and 71604 connections. For each fine connection, the file records how many subjects exhibit that connection, together with aggregate statistics about its fiber count and length. As a first preprocessing step, we retain only fine connections occurring in at least five subjects. This reduces the number of edges from 71604 to 37477, while leaving all 1015 parcels in the graph. The five-subject threshold is not explicitly stated in [9]; rather, it is inferred from the public dataset as the unique integer threshold reproducing the 37477 fine edges reported in the reference. For comparison, thresholds of four and six subjects yield 40895 and 34718 edges, respectively. The resulting fine graph is then mapped onto the 83 anatomical regions of the FreeSurfer parcellation, as in [9]. Several fine parcels therefore collapse onto the same anatomical region. Among the 37477 retained fine connections, 7895 connect parcels belonging to the same final region and consequently become internal to an aggregated node. The remaining 29582 connections join distinct regions, but many connect different fine parcels belonging to the same pair of anatomical regions; so, such parallel connections are merged, resulting in the final graph with 83 vertices and 1130 edges.

Each retained fine connection f is assigned the weight $w_f = \frac{n_f}{<\ell_f>}$, where n_f and $< \ell_f >$ denote its statistically aggregated values of fiber count and length, respectively. When several fine connections join the same pair of anatomical regions I and J, their contributions are summed, resulting in the final weight A_{IJ} of a bundle.
Before any simulation is performed, the reconstructed connectome is verified against the graph statistics reported in [9]. Our results are reported in Table 4[1].

Quantity	Value
fine graph vertices / edges	1015 / 37477
aggregated vertices / edges	83 / 1130
degree range	6–48
mean fibre number	40.16
mean fibre length	38.40 mm
mean adjacency $\bar{A}_{IJ}$	1.5702
adjacency range	0.0085–35.3221
mean weighted degree $\bar{D}_{II}$	42.7547
weighted degree range	2.0505–127.6435

Table 4: Statistics of the reconstructed 83-region connectome. The degree of a region is the number of regions it connects to and its weighted degree $D_{II} = \sum_J A_{IJ}$ the total connectivity it carries. The fibre number of a connection is the number of fibre tracks that tractography reconstructs between its two regions, as a median over the cohort: the same holds for the fibre length. Here their average values over the connections are reported. Rounded to the precision at which [9] prints them, all the values coincide with the published ones.

The definition of the connectivity weight has an important consequence for the resulting network. Since each fine-edge contribution is proportional to a fiber count and inversely proportional to a fiber length, connections supported by many short fibers receive larger weights, whereas long-range connections are penalized. After aggregation, this effect is inherited by the regional adjacency A_{IJ} through the sum of all fine connections joining the same pair of regions. At the level of the anatomical groups used in this chapter, the total connectivity between the frontal and parietal groups is 174.8 and that between the parietal and occipital groups is 189.7. In contrast, the total frontal-temporal connectivity is only 0.85. Figures 19, 20 and 21 show the resulting computational domain.

Figure 22 summarizes the connectivity of the reconstructed graph through the degree of its vertices, their weighted degree and the adjacency matrix. The latter reflects the same organization described in [9], as we're going to explain. The two dense diagonal blocks correspond to connections within the two hemispheres, whereas the off-diagonal blocks are much sparser and contain the interhemispheric connections. There are strong connections also within the four lobes, indicated along the diagonal of the matrix, drawn from the bottom-left to the top-right corner (4 lobes × 2 hemispheres). The coloured strips along the four sides identify the seven anatomical groups used in [8]. Neither these groups nor the four cortical lobes labels are part of the Budapest dataset, whose vertices carry their FreeSurfer anatomical names, the hemisphere and a cortical or subcortical flag, but no such anatomical grouping: both partitions (the four lobes classification of [9] and the seven classes of [8]) are obtained from the dataset anatomical names, through the name-based classification rules implemented in the solver. We resorted, in particular, to a `classify` function, which follows the standard Desikan–Killiany anatomy. We also observe that although the vertices retain their anatomical interpretation as identifiable brain regions, the coarse-grained edges do not possess a unique physical length after aggregation, since each of them represents the combined effect of several tractographic connections produced by the preprocessing rather than a single anatomical tract with a well-defined geometric length. We therefore assign a unit metric length, $h_e = 1$, to every edge and encode the tractographic information through the corresponding connectivity weight. The non-zero adjacency values span $0.0085 \leq A_{IJ} \leq 35.3221$, with a mean of 1.5702, reproducing the published range and mean to the reported precision. Within Fig. 22, panel (c), the most

[1] Although [9] describes the edge weights in terms of fiber quantities averaged over the patients' cohort, the published fine-graph statistics are reproduced by using the *median* fiber-count and length fields of the public dataset. Specifically, the reconstruction uses $w_f = \frac{n_f^{\text{median}}}{(<\ell_f>)^{\text{median}}}$, followed by summation over parallel fine connections for the construction of the final graph. We therefore retain the median-based preprocessing, while treating this choice as a data-driven reconstruction rather than as a statement about the undocumented implementation used in the reference.

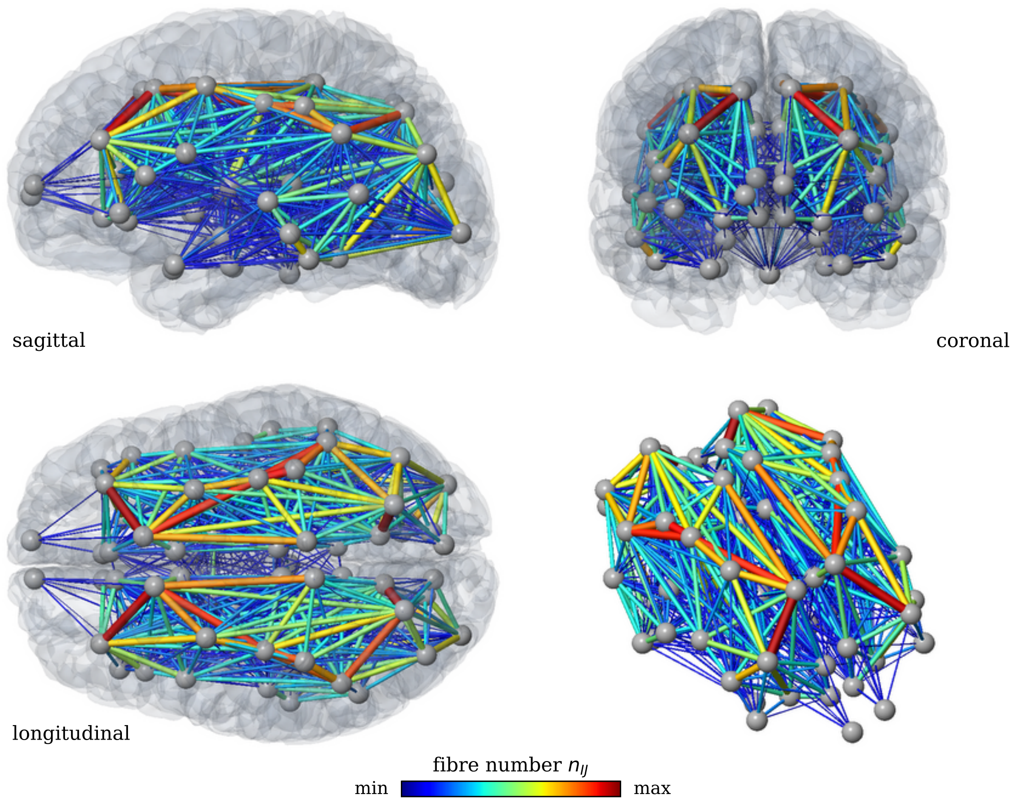


Figure 19: The connectome as a brain network: sagittal, coronal, longitudinal and oblique views of the 83 vertices and 1130 connections with and without the translucent pial surface. Colour and thickness of a connection both grow with its fibre number, from 1 to 595.5.

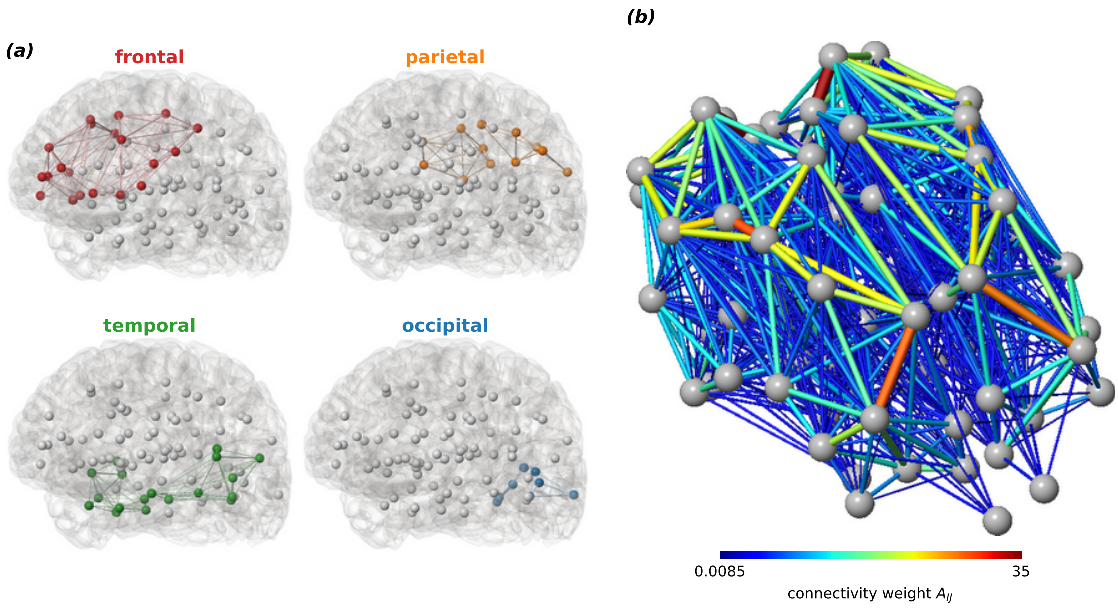


Figure 20: (a) The four cortical lobes: the regions of a lobe are coloured and the connections internal to it are drawn; white vertices belong to the remaining regions of a finer classification. (b) The resulting graph in the oblique view without the pial surface, connections coloured by connectivity weight. This is the abstract object the finite element method computes on; since an aggregated connection has no unique physical length, we choose for every edge a unit metric length: the anatomical information enters through its weight, the vertex coordinates being used for visualization only.

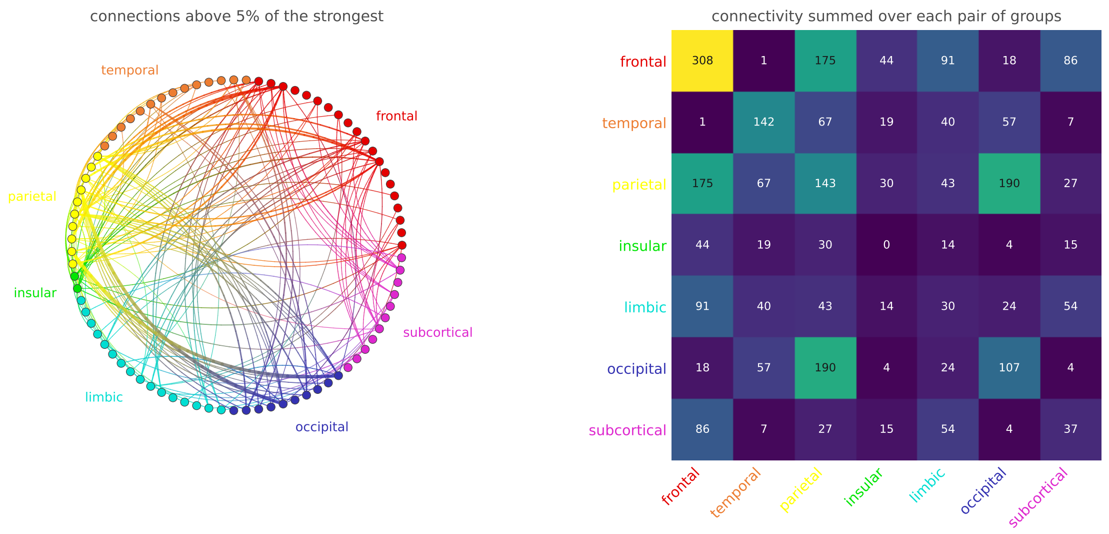


Figure 21: Connectivity of the reconstructed graph at the level of the seven anatomical groups. Left: the connections between regions of different groups with a weight above 5% of the strongest one, on a circular layout grouped by anatomical group, following [8]; each arc blends the colours of its two groups. Right: total connectivity between each pair of groups, measured as the sum of the weights A_{IJ} of all the connections joining a region of one group to a region of the other, the diagonal holding the connectivity internal to each group.

important feature that can be extracted is the pronounced heterogeneity of the network: most pairs of regions are either disconnected or only weakly connected, whereas a comparatively small number of connections carry much larger weights. Together with the heterogeneous weighted degree in panel (b), this reveals the presence of strongly connected hubs embedded in a much more weakly connected network. This structure is directly relevant to the spread of the disease. In the network model of [9], highly connected regions are expected to become effective hubs for the propagation of misfolded proteins: the disease does not propagate over a uniform network, but through a small number of preferentially connected pathways and hubs.

4.3 Network model and metric-graph formulation

We solve the Fisher–Kolmogorov equation posed on the (metric) graph presented in the previous section; on every edge e:

$$\partial_t c(s,t) - \partial_s[D_e\, \partial_s c(s,t)] = \alpha(s)\, c(s,t)\, [1 - c(s,t)]. \tag{10}$$

Up to the nonlinear right-hand side, this is the same parabolic problem studied in the first part of this work. The only new ingredient is the nonlinear reaction term. The references solve a simpler system, with one unknown per node and no spatial structure inside the connections. We will refer to the models of the references [8, 9] as the *nodal models*, that serve as the term of comparison next to our finite element solution. Each reproduced experiment uses the time scheme and way of treating the nonlinear term of its own reference. Table 5 summarizes the two schemes.

	Fully implicit [7, 9]	Semi-implicit [8]
Time discretization	Backward Euler ($\theta = 1$)	Crank–Nicolson ($\theta = 1/2$)
Nonlinear term	implicit, at t^{n+1}	semi-implicit, second-order extrapolation
Initialization	$\mathbf{c}^0$	$\mathbf{c}^{-1} = \mathbf{c}^0, \mathbf{c}^0$
Solve per step	Newton iteration	linear system, matrix rebuilt
Order in time	first	second

Table 5: The time discretization strategies adopted by the references, as implemented in this work. Both are based on the θ-method, combined with a different treatment of the nonlinear reaction term. In the semi-implicit scheme, each time step requires only a linear solve. The resulting matrix must however be rebuilt and refactorized at every step.

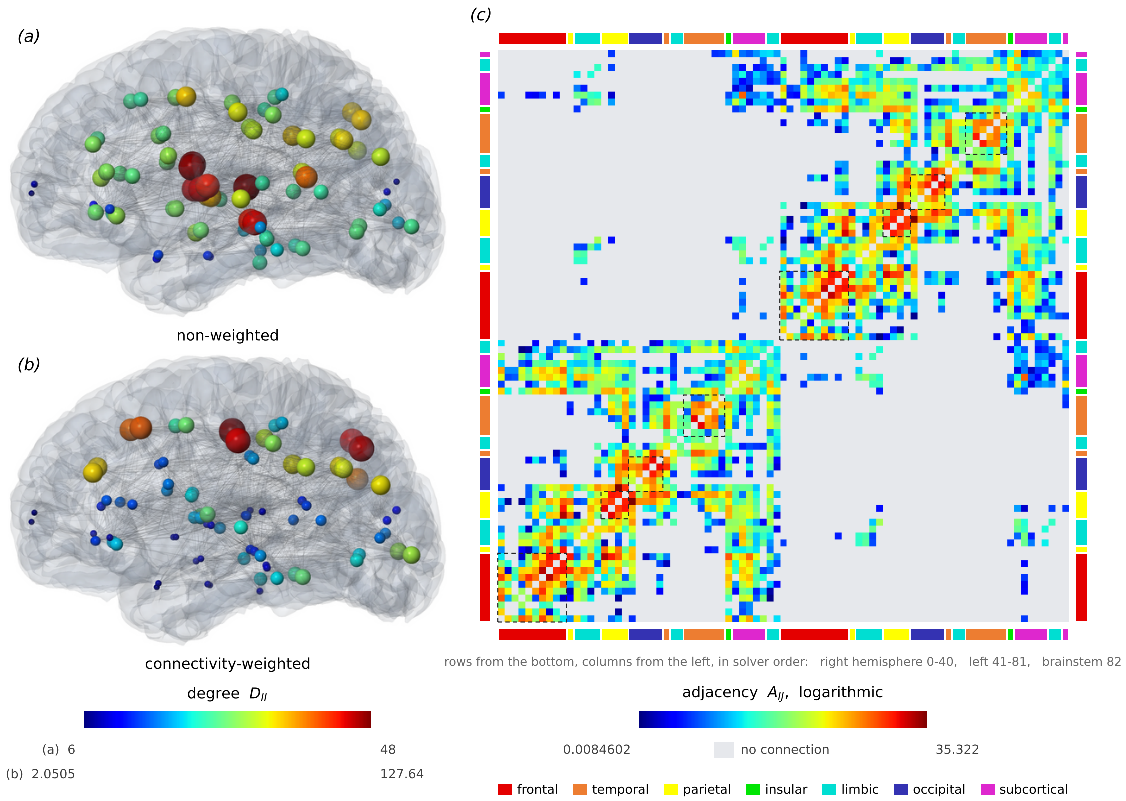


Figure 22: (a) Non-weighted degree and (b) connectivity-weighted degree; each panel is normalized to its own range. (c) Weighted adjacency matrix A_{IJ} displayed on a base-10 logarithmic scale over the complete non-zero range. A logarithmic colour scale is used to make both weak and strong connections visible within the same panel. Missing connections are shown separately in grey, while the coloured strips identify the seven anatomical groups. Rows and columns follow the solver ordering, with rows displayed from the bottom: right hemisphere 0–40, left hemisphere 41–81 and brainstem 82. The comparison of (a) and (b) shows that the regions with the most connections are not necessarily the hubs of the weighted network.

4.4 Comparison with the network model

We reproduce the baseline simulation of [9]: conversion rate $\alpha = 0.5$, misfolded protein seeded in the entorhinal cortex at $c_0 = 0.1$, all other regions healthy and 100 implicit steps of $\Delta t = 0.4$ years. Following the reference, the evolution is summarized by a biomarker: for a group of regions $\mathcal{I}$, the abnormality is the concentration averaged over its vertices and expressed as a percentage,

$$C_{\mathcal{I}}(t) = \frac{100}{|\mathcal{I}|} \sum_{I \in \mathcal{I}} c_I(t), \tag{11}$$

evaluated for the four cortical lobes and for the network as a whole. The simulation is run both with the nodal model, the formulation of the reference implemented literally and with the finite element model, here at one element per connection, so that the two systems share the same 83 unknowns and the same diffusion matrix and differ in the formulation alone. The biomarker curves have the expected sigmoid shape in both. The 50-percent crossing of the network average occurs at 11.11 years for the nodal model and at 12.68 years for the finite element model. The shift is expected: in the nodal model the domain consists of the 83 vertices only, whereas in the finite element model the connections are part of the domain as well and they are initially uninfected. The same seed, $c_0 = 0.1$ on the two entorhinal vertices, must therefore also fill the metric space between the vertices before their concentrations rise and the finite element curve shifts to the right by about 1.5 years. The two formulations describe different discrete systems, so their absolute crossing times are not expected to coincide. The activation sequence, however, is not reproduced by either: [9] report a separation of roughly five years between the temporal and the occipital lobe; our four lobe curves cross the 50-percent level within $3.9 \cdot 10^{-7}$ years of each other in the nodal model and $8.4 \cdot 10^{-3}$ years in the finite element model. The following section examines this discrepancy.

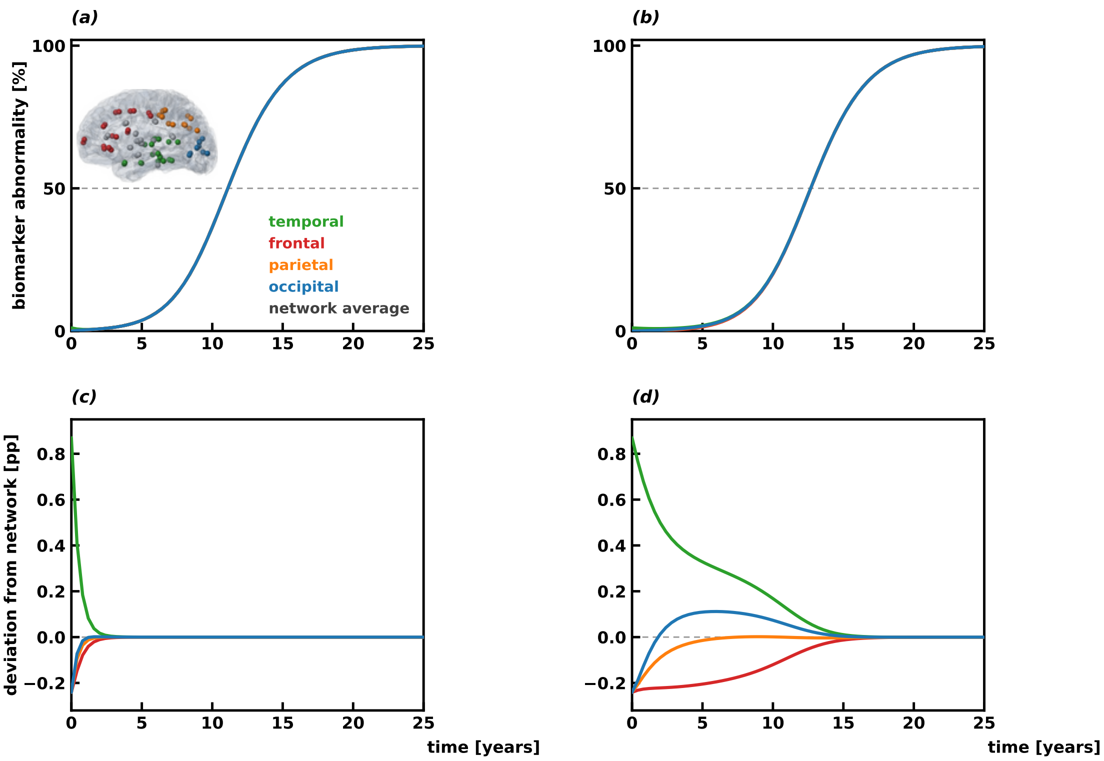


Figure 23: Biomarker abnormality of the four lobes and of the network, at $\alpha = 0.5$ with 100 steps of $\Delta t = 0.4$ years: (a) nodal reference, (b) metric-graph finite element model (one element per connection), with the four lobes in the colours of the inset. In both models the biomarker is the average of the concentrations at the 83 anatomical vertices, so the two curves start from the same value; the simulation runs to 40 years and the first 25 are shown, the curves having saturated by then. The four regional curves nearly coincide, crossing 50% within $3.9 \cdot 10^{-7}$ years of each other in (a) and $8.4 \cdot 10^{-3}$ years in (b), so the lower panels (c) and (d) report each curve minus the network average.

4.5 What controls the regional separation

Before explaining the discrepancy, it is useful to make some observations.
First, the graph. Although the reconstructed connectome reproduces the topology and global statistics reported in the reference, local differences in individual edge weights remain entirely possible, since part of the preprocessing pipeline had to be inferred from the public data rather than reproduced from a documented procedure.
Second, the discrepancy is not caused by the numerical model (as seen in the previous section) or by the definition of the lobes. In particular, the near-synchronization is already present at the level of the 83 individual regions: the biomarker concentrations all cross the 50% level within $6.4 \cdot 10^{-5}$ years in the nodal model and within 0.20 years in the finite element formulation, so no grouping of the nodes into lobes can separate the lobe curves by years. Nor is the reference's separation recovered by reasonable changes of the space and time discretization rules. Every tested combination reported in the repository show that these changes move the absolute timing of the curves, but not their near-synchronization.
Third, some implementation details remain unspecified: not only the complete graph-aggregation procedure, but also the precise implicit scheme adopted and the effective temporal scaling associated with the diffusion term. The reference defines the connectivity weights as $A_{IJ} = \frac{n_{IJ}}{<\ell_{IJ}>}$, where the fibre count (n_{IJ}) is dimensionless and the fibre length (ℓ_{IJ}) is measured in millimetres. Therefore, if these quantities are interpreted dimensionally, $[A_{IJ}] = \mathrm{mm}^{-1}$. On the other hand, the evolution equation is expressed in years, so that $\left[\frac{dc}{dt}\right] = \mathrm{yr}^{-1}$. The reaction term is dimensionally consistent because the conversion rate satisfies $[\alpha] = \mathrm{yr}^{-1}$. For the transport term, however, dimensional consistency would require an additional coefficient, say ρ, such that

$$\frac{dc}{dt} + \rho L c = \alpha\, c(1-c),$$

with

$$[\rho] = \frac{\mathrm{mm}}{\mathrm{yr}},$$

so that ($[\rho L] = \mathrm{yr}^{-1}$). In this interpretation, ρ would set the temporal scale of transport on the network, analogously to how α sets the time scale of the reaction. Such a transport coefficient is not introduced explicitly in the published formulation. We noticed that the relative strength of network transport and local reaction strongly affects the temporal separation between the lobes. With the reaction switched off, the nodal model and the finite element model read

$$\frac{d\mathbf{c}}{dt} = -\rho\, L\, \mathbf{c} \qquad \text{and} \qquad \mathbf{M}\,\frac{d\mathbf{c}}{dt} = -\rho\, L\, \mathbf{c},$$

where L is the same matrix in the two, because at one element per connection the discrete Hamiltonian of Section 2.2 with $V = 0$ and diffusion coefficients $D_e = w_e$ coincides with the connectome Laplacian. What changes is the matrix in front of the time derivative.

In both cases the solution is a sum of exponentially decaying modes. The mode with zero rate is the uniform state the network tends to; every other mode decays at the rate $\rho\lambda_k$, where the λ_k are the eigenvalues of L in the nodal model and the eigenvalues of L relative to $\mathbf{M}$, $L\mathbf{v} = \lambda\,\mathbf{M}\mathbf{v}$, in the finite element one. The slowest non-zero rate decides how long a difference between regions can survive, so diffusion homogenizes the network on the time scale

$$\tau_{\text{diffusion}} \sim \frac{1}{\rho\lambda_2},$$

with λ_2 the smallest non-zero eigenvalue of the model at hand. Reaction alone makes the concentration grow at every vertex on the time scale

$$\tau_{\text{reaction}} \sim \frac{1}{\alpha}.$$

The two processes compete and their ratio defines the Damköhler number

$$\mathrm{Da} = \frac{\tau_{\text{diffusion}}}{\tau_{\text{reaction}}} = \frac{\alpha}{\rho\,\lambda_2}. \tag{12}$$

When Da $\ll 1$ the network homogenizes faster than the reaction grows and strong regional synchronization is expected; when Da is large the differences created by the seed survive and the lobes' abnormality curves can separate.

For the reconstructed connectome the nodal model has $\lambda_2 = 0.7723$. In the finite element model, the same diffusive term $\rho L\mathbf{c}$ is filtered through the mass matrix $\mathbf{M}$, so the rate of change is governed by $\mathbf{M}^{-1}L$ rather than by L alone. With mass lumping (see Section 4.5.1 for the description of the mass lumping as a stabilization technique) the generalized spectrum changes again; in this connectome the second eigenvalue decreases from 0.0635 (the value of the consistent mass) to 0.0606, so the slowest diffusive mode becomes slightly slower. This decrease is not obvious a priori and has to be determined from the generalized eigenvalue problem. So, in the finite element model, the smallest non-zero eigenvalue is twelve to thirteen times smaller than 0.7723. With $\rho = 1$ and $\alpha = 0.5$, the nodal model therefore has $\mathrm{Da} = 0.65$, while the finite element model has $\mathrm{Da} = 7.9$. At this point, however, the reasoning seems to break down: despite $\mathrm{Da} = 7.9$, the lobe-wise curves remain nearly synchronized. This suggests that the slowest diffusive mode of the whole network is not necessarily the one that controls the separation between lobes. For this reason, we also consider the eigenvectors associated with the spectrum containing λ_2. Each eigenvector describes a different spatial pattern of concentration differences that diffusion tends to smooth out, while the corresponding eigenvalue determines how fast that pattern decays. We find that the mode associated with λ_2 does not represent a contrast between anatomical lobes. Therefore, for the lobe-wise biomarker, the relevant quantity is not the slowest eigenvalue of the entire network, but the slowest relaxation rate among the modes that actually describe differences between lobes.

The eigenvector associated with λ_2 confirms this: its largest components are concentrated on the frontal and temporal poles and on the entorhinal cortices, with opposite signs in the two hemispheres. Therefore, the slowest diffusive mode of the whole network represents mainly an inter-hemispheric peripheral imbalance rather than a contrast between anatomical lobes.

The decay rates, for the nodal model, range from 10.0 to 49.4, with the temporal-occipital contrast, for instance, relaxing at a rate of 21.6. For the finite element model, the slowest decay rate is 1.11 with the consistent mass matrix and 0.86 with the lumped version. With these rates, we can define a lobe-scale Damköhler number,

$$\mathrm{Da}_{\mathrm{lobe}} = \frac{\alpha}{\rho\,\lambda_{\mathrm{lobe}}}.$$

At $\rho = 1$, it is 0.05 for the nodal model and 0.45 and 0.58 for the finite element model, with the consistent and the diagonalized mass matrix respectively. In every case we have a $\mathrm{Da}_{\mathrm{lobe}} < 1$, so at the lobe-scale, transport still beats reaction and the synchronized curves are exactly what this number predicts. Figures 24 and 25 quantitatively confirm the reading: the curves' separation becomes appreciable as $\mathrm{Da}_{\mathrm{lobe}}$ crosses one.

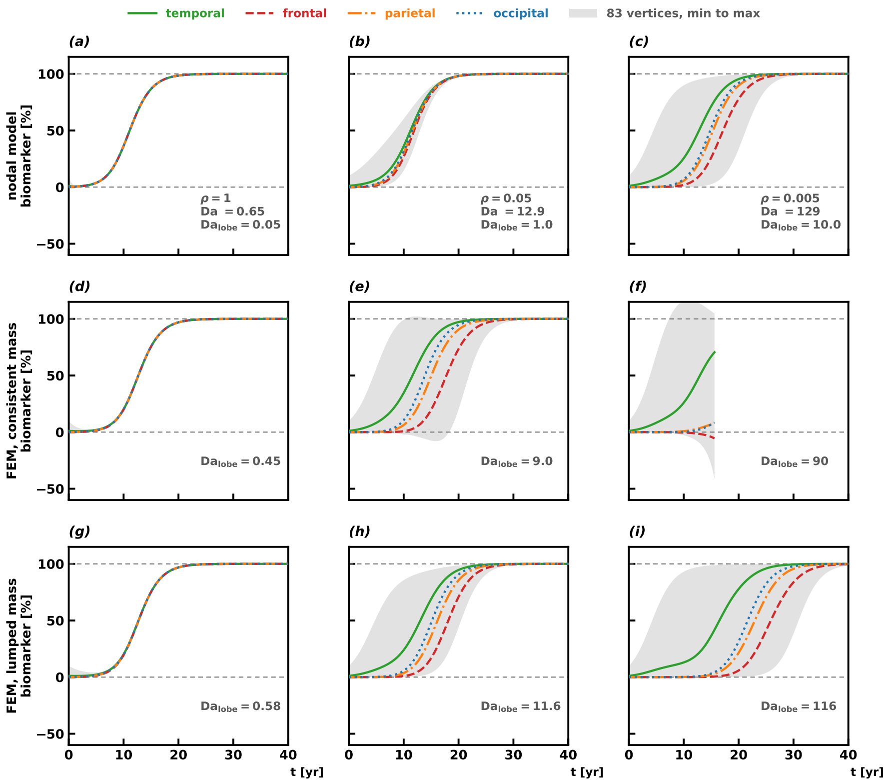


Figure 24: Regional separation as a function of the Damköhler number (12), for the three models, at one element per connection with the fully implicit scheme, $\Delta t = 0.4$ years, the entorhinal seed and $\rho = 1$, 0.05 and 0.005 (columns). Rows: (a)–(c) the nodal model of the references; (d)–(f) the finite element model with the consistent mass matrix of the first part of this work; (g)–(i) the same model with the lumped mass matrix (see Section 4.5.1). Each panel shows the four lobe biomarkers and, shaded, the envelope of the 83 vertex concentrations; the dashed lines mark 0 and 100 percent. The lobes separate as Da grows in every model. With the consistent mass, the envelope leaves the physical range at Da = 12.9 (e) and at Da = 129 (f), where the Newton iteration fails at $t = 15.6$ years and so the run stops. With the lumped mass, the solution stays within $[0, 1]$ in every panel.

Since the conversion rate is uniform, the different lobe activation times cannot be explained by α and are actually determined by the position of the seed (here at the temporal lobe) and by the network connectivity. The total temporal connectivity, in particular, suggests the ordering parietal–occipital–frontal, with weights 75.0, 55.6 and 1.0, respectively. However, the biomarker is averaged over all regions of each lobe. Once the connectivity is considered per region, the temporal-occipital coupling is stronger than the temporal-parietal one (7.0 versus 6.3 on average). The resulting ordering is therefore temporal–occipital–parietal–frontal.
We conclude that:

- The relative strength of reaction and transport, quantified by the Damköhler analysis, controls *how much* the lobe curves separate in time (see panel (b) of Figure 25).
- For a fixed seed and a uniform conversion rate, the spatial distribution of the connections of the reconstructed connectome controls *in which order* the lobes cross the biomarker threshold.

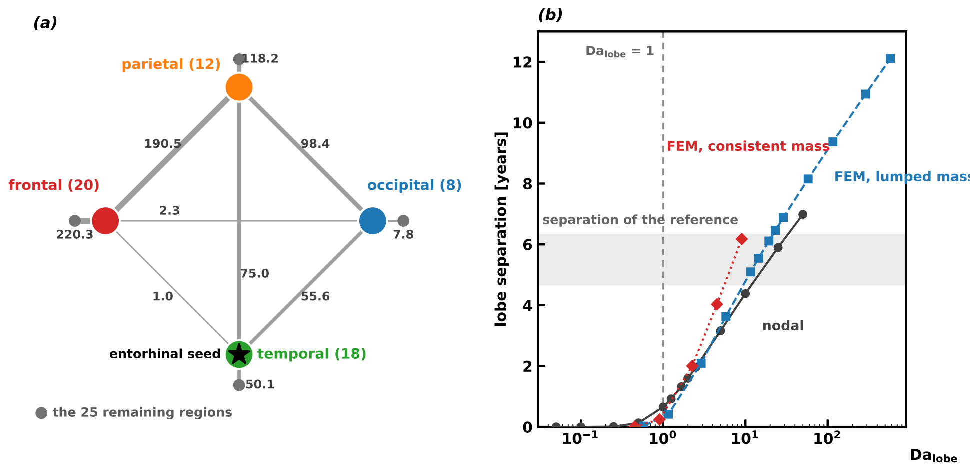


Figure 25: (a) The connectome compressed over the four lobes of the biomarker. (b) Lobe separation against $\mathrm{Da}_{\mathrm{lobe}}$ for the three models; the dashed line marks $\mathrm{Da}_{\mathrm{lobe}} = 1$, below which every model keeps the lobes together and the grey band is the separation of the reference, 5.5 years read from figure 7 of [9] with a margin of 15%.

4.5.1 Mass lumping as a stabilization technique

With the finite element model, using one element per connection and the fully implicit time scheme, we observe that at low and moderate Damköhler numbers the solution may temporarily leave the admissible interval $[0, 1]$, but it eventually recovers. At $\mathrm{Da} = 0.65$ the minimum concentration reaches only $-1.3 \cdot 10^{-4}$, while at $\mathrm{Da} = 12.9$ it reaches -0.08 and the maximum reaches 1.02. From the next tested scaling, $\mathrm{Da} = 16.2$, the solution no longer recovers. At $\mathrm{Da} = 129$ the concentrations reach -0.41 and 1.17 by $t = 15.6$ years, after which the nonlinear solver cannot complete the next time step. The same qualitative behaviour is observed with the semi-implicit scheme of Table 5 and the failure persists under mesh refinement, indicating that it is not simply a coarse-mesh effect.
The origin of this behaviour can be understood from the finite element system

$$\mathbf{M}\frac{\mathbf{c}^n - \mathbf{c}^{n-1}}{\Delta t} + \mathbf{H}\mathbf{c}^n = \mathbf{f}(\mathbf{c}^n),$$

where $\mathbf{H}$ is the discrete Hamiltonian of Section 2.2 with $V = 0$ and diffusion coefficients $D_e = \rho w_e$, so that $\mathbf{H} = \rho L$ at one element per connection. When transport is weak, the diffusion term becomes small and the update is approximately governed by

$$\mathbf{c}^n - \mathbf{c}^{n-1} \approx \Delta t\, \mathbf{M}^{-1}\mathbf{f}(\mathbf{c}^n).$$

With the consistent mass matrix, $\mathbf{M}^{-1}$ contains negative off-diagonal entries. This means that, through $\mathbf{M}^{-1}$, the reaction computed at one vertex can contribute with the opposite sign to the increment of neighbouring vertices, producing small negative concentrations. Once a concentration crosses below zero, the logistic term reinforces the error because

$$c < 0 \qquad \Longrightarrow \qquad c(1 - c) < 0.$$

The instability therefore develops in two stages: the consistent mass discretization first generates an undershoot and the nonlinear reaction then amplifies it. When transport is strong, diffusion smooths these small excursions and the solution returns to the physical interval; when transport becomes too weak, it can no longer compensate for them.
The Newton failure observed at $\mathrm{Da} = 129$ is therefore a consequence rather than the cause of the instability. By $t = 15.6$ years the solution has already reached $c_{\min} \simeq -0.41$ and the logistic reaction tends to drive the negative concentrations further down. Since the reaction is evaluated at the new time level, the fully implicit scheme requires the solution of a nonlinear system at every time step. Newton's method solves this system iteratively by correcting a current approximation until the residual is sufficiently small. In our implementation, the Newton iteration is damped and equipped with a safeguard that prevents the intermediate iterates from leaving the interval $[-0.5, 1.5]$. At the step following $t = 15.6$ years, the corrections required by the nonlinear system tend to push some

concentrations below the lower bound. The damping can reduce these corrections, but eventually the solver cannot both remain inside the safeguard and reduce the residual to the required tolerance, so the iteration stops. The semi-implicit scheme, which solves a linear system at every step and has no such safeguard, continues at the same scaling and eventually diverges to $[-507, 523]$. This confirms that Newton is not creating the instability: it is only the point at which the fully implicit computation can no longer follow an already nonphysical solution.

The same loss of positivity is reproduced in the one-dimensional problem of Section 4.1, showing that the mechanism is not specific to the connectome topology.

A standard remedy for this loss of positivity is *mass lumping*. The consistent mass matrix is replaced by the diagonal matrix obtained from its row sums, while the reaction integral is evaluated using the vertex rule[2],

$$f_i = M_{ii}^L \, \alpha_i c_i (1 - c_i).$$

Together, mass lumping and the vertex rule make the reaction treatment local: the lumped mass removes the cross-node coupling introduced by $\mathbf{M}^{-1}$, while the vertex rule makes f_i depend only on the local concentration c_i. The reaction therefore takes the same node-wise form as in the nodal model, while the transport operator remains the finite element one.

We added mass lumping as an option in the finite element library and repeated the sweep under the same conditions. With the consistent mass, the concentration envelope already leaves $[0, 1]$ at $\mathrm{Da} = 12.9$ and the computation eventually breaks down at $\mathrm{Da} = 129$. With the lumped mass, instead, all vertex concentrations remain within $[0, 1]$ throughout the sweep, while the relevant physical behaviour is preserved: the lobe curves progressively separate as transport is weakened, as discussed in the previous paragraph.

At the original transport scale, the two formulations are almost indistinguishable, with the network crossing time changing only from 12.68 to 12.72 years. This shows that mass lumping stabilizes the finite element model in the weak-transport regime without significantly modifying the solution where the consistent formulation is already reliable. For this reason, in the rest of the chapter we use the consistent mass where transport is strong and the lumped mass where transport is weak, as in the seeding experiment of Section 4.6. Every result from this point onward is therefore a finite element result.

4.6 Tau seeding and expected progression

The two hallmark proteins of Alzheimer's disease are tau and amyloid-β. Amyloid-β is not studied here: the connectome encodes axonal pathways, which are the propagation routes of intracellular tau, whereas amyloid-β spreads extracellularly, which is why the reference itself simulates only tau on this graph [9]. Tau enters the model through the initial spatial location alone: inclusions appear first in the transentorhinal region and the expected progression is from the entorhinal cortex to the temporal lobe and then to the interconnected neocortex [7]. On the graph, the seeding is the entorhinal vertices at $c_0 = 0.1$.

Figure 26 shows the computed progression below its clinical counterpart, namely the staging drawings of [7]. Three representative stages are selected as the first instants at which the network mean concentration reaches 10%, 40% and 80%. The solution is the finite element one with lumped mass and the fully implicit scheme, at $\rho = 0.005$, corresponding to $\mathrm{Da} = 129$ and $\mathrm{Da}_{\mathrm{lobe}} = 116$.

This transport scaling is essential for the progression to become visible. At $\rho = 1$, the scale used in the references, transport is sufficiently fast to synchronize the network: in a control simulation the four lobes cross the 50% level within 0.03 years of each other and the three stages would appear almost spatially uniform. The choice $\rho = 0.005$ should therefore be interpreted as a calibration of the transport time scale against the reference progression, rather than as an additional biological input. The only biological input of the simulation is then the location of the initial seed.

[2] The vertex rule approximates an integral over an edge $e = [i, j]$ of length h_e using only the values at its endpoints. Applied to the reaction term,

$$\int_e \alpha \, c_h (1 - c_h) \phi_i \, \mathrm{d}s \approx \frac{h_e}{2} \left[\alpha c_i (1 - c_i) \phi_i(i) + \alpha c_j (1 - c_j) \phi_i(j) \right].$$

Since $\phi_i(i) = 1$ and $\phi_i(j) = 0$, this reduces to

$$\frac{h_e}{2} \alpha c_i (1 - c_i).$$

After summing over all edges incident to vertex i, one obtains $f_i = M_{ii}^L \alpha c_i (1 - c_i)$, so the reaction at vertex i depends only on the local concentration c_i.

What Figure 26 recovers has to be stated carefully, because the first station of the expected pattern is prescribed by the seed rather than produced by the model and because every run of Eq. (10) with a positive reaction rate ends with the whole network at $c = 1$. For the tau protein, the model produces one step of the expected sequence: from the entorhinal seed the pathology reaches the temporal lobe first, as the clinical description says. It does not produce the rest. The remaining lobes follow in the order of the graph-connectivity, so the occipital lobe, which the sequence of [9] leaves for last, is here the second to be involved and the frontal lobe, which the reference places immediately after the temporal one, is here the last, according to our Figure 24. In summary, the computed row starts from the entorhinal seed and fills the temporal lobe first; the spreading then moves posteriorly, the occipital lobe being the second of the four to cross the 50% level, before progressively involving the rest of the network.
The same caveat discussed in Section 4.5 remains valid. Within the later tau stages, the relative ordering of the cortical lobes is determined by the connectivity of the reconstructed graph rather than by the clinical staging sequence. The four lobes cross the 50% level in the order temporal, occipital, parietal and frontal, reproducing the ordering already observed in Figure 24. In our simulations, the frontal group is delayed by its weak effective connectivity, with the frontal pole being the least connected region of the graph. Since the conversion rate is uniform in the present model, regional differences in the progression are driven by connectivity alone. This motivates the introduction of region-dependent conversion rates, as in Section 4.7, where additional biological heterogeneity is included in the reaction term.

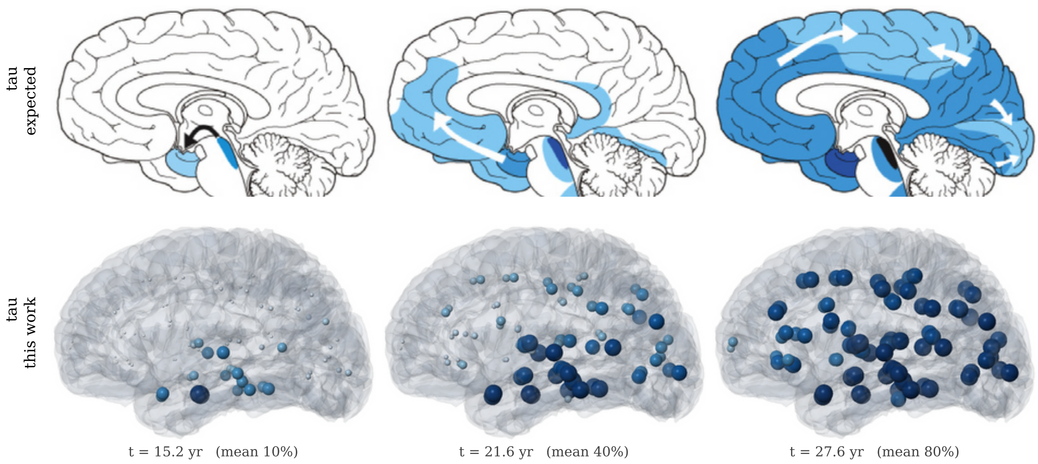


Figure 26: Expected and computed progression for tau. First row: clinical staging drawings from Figure 1 of [7], medial view. Second row: the progression computed in this section, with sphere radius and blue intensity growing with the local concentration, at the first instants at which the network mean reaches 10%, 40% and 80%. The finite element model uses lumped mass, $\rho = 0.005$, $\mathrm{Da} = 129$ and $\mathrm{Da}_{\mathrm{lobe}} = 116$. Tau, seeded at the entorhinal cortex, first involves the temporal neighbourhood and then progressively the rest of the network. The three stages highlight the origin, direction and endpoint of the progression, determined by the network connectivity, as discussed in the text. At the second computed stage the temporal lobe has reached 86% and the frontal lobe 13%, with the occipital and parietal lobes in between at 49% and 36%, so the computed row shows the connectivity order and not the clinical one, whose second station is the frontal lobe. This motivates the introduction of further biology through heterogeneous reaction rates.

4.7 Deterministic model with regional reaction rates

The previous sections showed what can be explained by the connectome alone. With a uniform conversion rate, the network connectivity strongly influences the ordering of the lobe-wise curves. However, this ordering does not fully reproduce the clinical one. We now introduce the biological heterogeneity that was absent from the previous model by allowing the conversion rate to vary between anatomical regions.
We follow [8], who solve a patient-specific stochastic Fisher–Kolmogorov model on a 246-node Brainnetome graph and estimate the reaction coefficient region by region from a PET acquisition. Their formulation is nodal, with one unknown per vertex and a discrete graph Laplacian, whereas here we solve the corresponding deterministic problem with the metric-graph finite element model developed in this work. We do not attempt to reproduce the clinical trajectory of their patient.

Instead, we transfer their estimated mean regional reaction rates to our 83-region connectome and ask whether they change the ordering of the four lobes. Making α region-dependent therefore introduces the first biological heterogeneity in the model beyond the connectivity itself.
The reaction coefficient is taken piecewise constant over seven anatomical groups, using the mean values reported in Table 1 of [8]:

$$\alpha_{\text{frontal}} = 0.1801, \qquad \alpha_{\text{temporal}} = 0.1421, \qquad \alpha_{\text{parietal}} = 0.0627,$$

$$\alpha_{\text{insular}} = 0.1005, \qquad \alpha_{\text{limbic}} = 0.1351, \qquad \alpha_{\text{occipital}} = 0.0545, \qquad \alpha_{\text{subcortical}} = 0.1147.$$

The experiment is the staging comparison of Section 4.6 repeated with these rates and nothing else changed: the same entorhinal seeding, the same scaling $\rho = 0.005$, the same lumped mass and the same three-stage rule, so that the solutions differ from the uniform ones only through the reaction coefficients. The seven rates are rescaled to the vertex mean 0.5 of the uniform run, which preserves the reaction budget, $\mathrm{Da} = 129$ and $\mathrm{Da}_{\text{lobe}} = 116$ and only redistributes the rate; the horizon is extended to 80 years for the slow groups. The rates of the insular, limbic and subcortical regions enter the model in the same way, so the dynamics of the four lobes is shaped also by what happens around them; the ordering we compare is the one of the four lobes alone, against the expected pattern of Section 4.6.
Figure 27 shows the outcome. The tau lobes now cross the 50% level at 14.7, 19.9, 29.7 and 32.4 years, in the order temporal, frontal, parietal and occipital, which is the clinical sequence the uniform rate could not produce; the lower panel shows the same run through the biomarker curves of the four lobes. The comparison therefore suggests a division of roles rather than a strict separation: connectivity governs the transport pathways and contributes to the ordering, while the heterogeneous reaction field provides the dominant regional bias in this experiment.

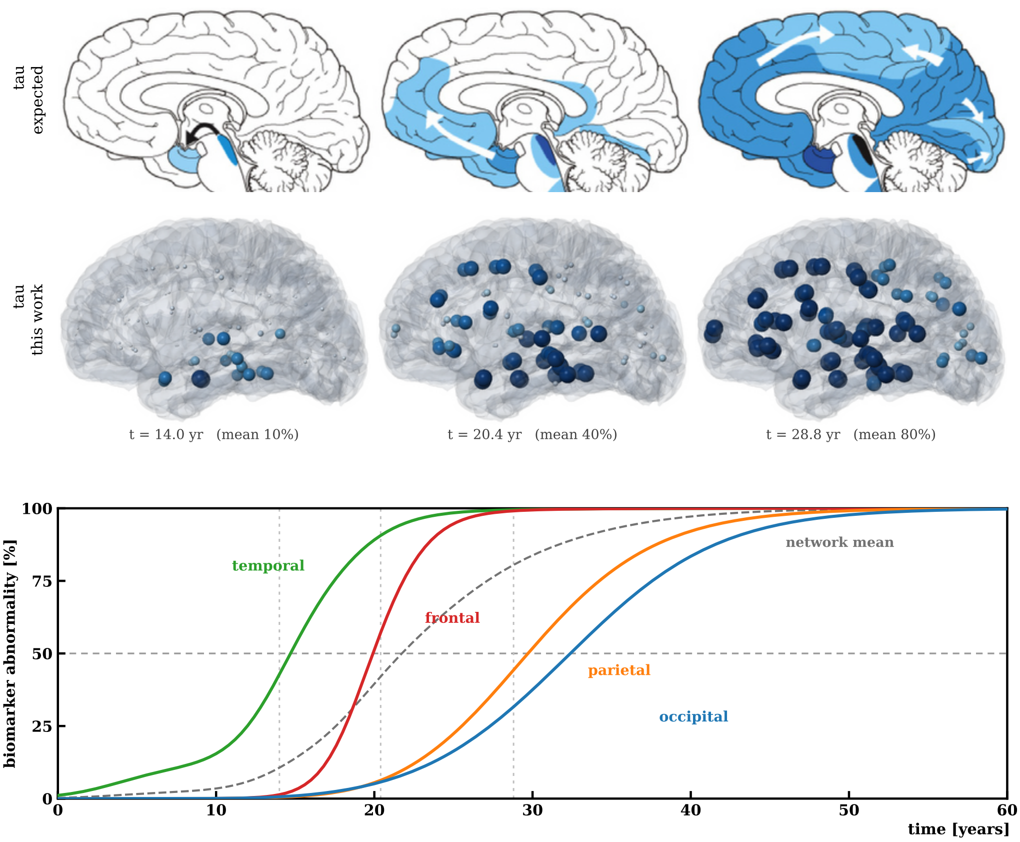


Figure 27: The experiment of Figure 26 repeated with the regional reaction rates of this section: same seed, same scaling $\rho = 0.005$, same lumped mass and the same three-stage rule, over 80 years. The computed disease progression now reaches the frontal lobe second and the occipital lobe last, which is the clinical sequence. Bottom row: the four lobe biomarkers of the same run, with the network mean dashed, the 50% level marked and the three stage instants as vertical lines, at 14.0, 20.4 and 28.8 years.

5 C++ implementation

5.1 Sequential baseline

The deterministic model above is also the reference problem for the performance analysis that opens the next chapter. Timings are taken in a separate release build with visualisation and diagnostic output disabled, thread counts pinned to one, and reported as the median of five runs after one warm-up.

Elements per connection	Unknowns	Matrix nonzeros	100 steps [s]	Per step [s]
1	83	2343	0.0832	0.00083
2	1213	5733	1.8694	0.01869
4	3473	12513	1.5486	0.01549
8	7993	26073	2.2343	0.02234

Table 6: Sequential baseline for the deterministic model. Assembly accounts for less than one percent of the total in every refined case.

Two facts emerge from Table 6 and they set the agenda for what follows. First, in every refined case more than 99% of the time is spent in the time loop (95.9% at one element per connection) and almost none in assembly or input, so the target for optimisation is the repeated sparse factorisation forced by the semi-implicit scheme (Table 5), not the finite element assembly. Second, the cost per step is not a function of the number of unknowns alone: the two-element mesh, with 1213 unknowns, costs more than the four-element mesh with 3473. The two-element mesh inherits a hub-dominated sparsity pattern from the connectome, and sparse factorisation depends on the elimination ordering and the resulting fill-in at least as much as on the matrix dimension.

For scale, [9] report that their nodal model runs its 100 implicit steps in 0.55 s without output on a standard laptop; our metric-graph formulation at the same 83 unknowns takes 0.083 s for its 100 semi-implicit steps. The machines and the schemes differ, so this is an order-of-magnitude check rather than a comparison, but it confirms that the P1 metric-graph formulation carries no cost penalty over the nodal network model at the coarsest mesh, while allowing arbitrary refinement inside the connections.

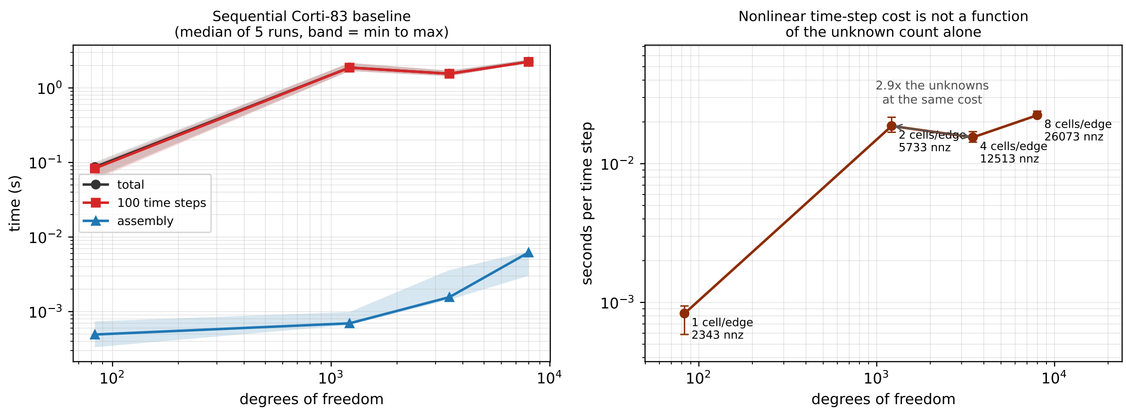


Figure 28: Sequential baseline. (a) assembly, time loop and total against the number of unknowns, with the band spanning the minimum and maximum of five runs. (b) cost per time step, showing that it is not a function of the unknown count alone: the two-element mesh costs more per step than the four-element one.

This is precisely the observation that motivates the reordering study of the next chapter, where bandwidth, profile, factor fill and factorisation time are measured for different orderings before any parallel scaling result is interpreted.

6 Conclusions

A Properties of the Combinatorial Laplacian

The combinatorial Laplacian $L_{\mathcal{G}} = MM^\top$ satisfies the following properties:

- Symmetry

$$L_{\mathcal{G}}^\top = (MM^\top)^\top = (M^\top)^\top M^\top = MM^\top = L_{\mathcal{G}}.$$

- Positive semidefiniteness

$$x^\top L_{\mathcal{G}}\, x = x^\top MM^\top x = (M^\top x)^\top (M^\top x) = \|M^\top x\|_2^2 \geq 0,$$

hence $x^\top L_{\mathcal{G}}\, x \geq 0$ for all $x \in \mathbb{R}^V$.

- Singularity

$$(M^\top f)_e = f(v_i) - f(v_j) \qquad \forall\, e = \{v_i, v_j\} \in \mathcal{E}, f \in \mathbb{R}^V$$

So $(M^\top f)_e = 0$ if and only if $f(v_i) = f(v_j)$ that is, f must be constant on every pair of adjacent vertices. If $\mathcal{G}$ is connected, this forces f to be globally constant, so

$$\ker(M^\top) = \operatorname{span}\{(1, 1, \ldots, 1)^\top\}.$$

Since $L_{\mathcal{G}} = MM^\top$, we have $L_{\mathcal{G}}\, f = 0 \iff M^\top f = 0$, hence

$$\ker(L_{\mathcal{G}}) = \ker(M^\top) = \operatorname{span}\{(1, 1, \ldots, 1)^\top\},$$

confirming that $L_{\mathcal{G}}$ is singular. More generally, the constant function argument applies independently on each connected component: a vector f satisfying $M^\top f = 0$ must be constant within each component but may take different values across components. Therefore

$$\dim \ker(L_{\mathcal{G}}) = k,$$

where k is the number of connected components of $\mathcal{G}$[3].

- Meaning

$$(Mg)_i = \sum_{e \ni v_i} \pm\, g_e, \forall g \in \mathbb{R}^E$$

where the sign is $+1$ if the flux along e is outgoing from v_i and -1 if it is incoming to v_i, reflecting the local flow configuration at its incident edges. This is the discrete analogue of a divergence: M accumulates the signed flux of g at each node.
Dually, for $f \in \mathbb{R}^V$, the action of $M^\top$ on edge $e = \{v_i, v_j\}$ reads

$$(M^\top f)_e = f(v_i) - f(v_j),$$

which is the difference of the nodal values at the endpoints of e, i.e. the discrete analogue of a derivative. Thus $M^\top$ plays the role of a *discrete gradient*.
Since M is a discrete divergence and $M^\top$ a discrete gradient, their composition $L_{\mathcal{G}} = MM^\top$ is naturally interpreted as a *discrete Laplacian*, which motivates calling $L_{\mathcal{G}}$ the Laplacian of the combinatorial graph $\mathcal{G}$.

Input graphs are specified in a simple `.graph` plain-text format: the first line gives $|\mathcal{V}|$ and $|\mathcal{E}|$, followed by one line per edge with endpoint indices and length.

[3]In a (possibly disconnected) graph, a connected component is defined as a maximal connected subgraph, i.e. a largest connected subset of vertices (together with the edges between them).

B Derivation of the weak and algebraic formulations

B.1 Problem formulation

A metric graph $\Gamma = (\mathcal{V}, \mathcal{E})$ consists of a finite set of vertices and a finite set of edges. The vertex set is generally partitioned into three disjoint subsets:

$$\mathcal{V} = \mathcal{T} \cup \mathcal{V}_1 \cup \mathcal{V}_2,$$

where $\mathcal{T}$ collects the vertices of degree one (i.e. they are the terminal points of the graph), $\mathcal{V}_1$ collects the *simple junction* vertices, at which the flux is conserved and $\mathcal{V}_2$ collects the *generalized junction* vertices at which a different local dynamics can be prescribed. On Γ we consider a system of parabolic problems. Each line of this system lives within an edge e and communicates with the others through boundary conditions.

For each edge $e \in \mathcal{E}$ and $t \in (0, T]$, the unknown $u_e(s,t) = u(s_e, t)$ satisfies

$$\frac{\partial u_e}{\partial t} - \frac{\partial}{\partial s}\left(\mu_e \frac{\partial u_e}{\partial s}\right) + r_e u_e = f_e, \qquad 0 < s_e < \ell_e, \quad \forall e \in \mathcal{E}, \tag{13}$$

where $\mu_e = \mu_e(s,t) > 0$ is the diffusion coefficient, $r_e = r_e(s,t) \geq 0$ is the reaction coefficient and $f_e = f_e(s,t)$ is a given source term on e. The initial condition on each edge is the following

$$u_e(s,0) = u_e^0(s), \qquad 0 < s < \ell_e \, , \forall e \in \mathcal{E}.$$

At each terminal vertex, either a Dirichlet boundary condition or a homogeneous Neumann condition can be imposed. At each simple junction vertex $v_p \in \mathcal{V}_1$, the solution is continuous and the total flux is conserved:

$$\sum_{e \in \mathcal{E}_{v_p}} \left[\mu_e \frac{\partial u_e}{\partial \mathbf{n}_e}\right]_{v_p} = 0, \qquad \forall \, v_p \in \mathcal{V}_1, \tag{14}$$

where $\mathcal{E}_{v_p}$ denotes the set of edges incident to v_p and $[\partial/\partial \mathbf{n}_e]_{v_p}$ denotes the outward normal derivative along e with respect to v_p. At each generalized junction vertex $v_q \in \mathcal{V}_2$, a local variable $u_q = u_q(t)$ is associated with the vertex and satisfies

$$c_q \frac{du_q}{\mathrm{d}t} + r_q u_q = \sum_{e \in \mathcal{E}_{v_q}} \left[\mu_e \frac{\partial u_e}{\partial \mathbf{n}_e}\right]_{v_q} + f_q, \qquad \forall \, v_q \in \mathcal{V}_2, \tag{15}$$

where $c_q > 0$ is a relaxation coefficient, $f_q = f_q(t)$ is a local source term, and $[\partial/\partial \mathbf{n}_e]_{v_q}$ is the outward normal derivative along each incident edge with respect to v_q. These are two different versions of the so-called Kirchhoff conditions.

B.2 Weak formulation

We consider the parabolic problem (13) on the metric graph $\Gamma = (\mathcal{V}, \mathcal{E})$, together with the Kirchhoff conditions (14)–(15), homogeneous Neumann boundary conditions at the terminal vertices and the global continuity constraints. The spatial differential operator acting on each edge is

$$\mathcal{L}_e \, u_e = -\frac{\partial}{\partial s}\left(\mu_e \frac{\partial u_e}{\partial s}\right) + r_e \, u_e, \tag{16}$$

so that (13) reads $\partial_t u_e + \mathcal{L}_e u_e = f_e$ on each $e \in \mathcal{E}$. Note that when μ_e is constant and $r_e = V_e(s)$ is a potential, $\mathcal{L}_e$ reduces to the quantum-graph Hamiltonian $\mathcal{H}_e u_e = -\mu_e \frac{\partial^2 u_e}{\partial s^2} + v_e u_e$.
The weak formulation is obtained by multiplying (13) by a test function $w \in H^1(\Gamma)$, integrating over each edge, summing over $\mathcal{E}$ and integrating by parts the diffusion term. At each junction vertex, the boundary terms arising from integration by parts are replaced using the Kirchhoff conditions (14)–(15), which cancel for $v_p \in \mathcal{V}_1$ and contribute a local ODE term for $v_q \in \mathcal{V}_2$. Also contributions coming from terminal vertices cancel, as we consider homogeneous Neumann conditions. The weak problem reads:

Find $u(s,t) \in H^1(\Gamma)$ such that

$$\sum_{e\in\mathcal{E}} \int_e \left[\frac{\partial u_e}{\partial t}\, w_e\right] \mathrm{d}s + a(u,w) - \sum_{v_q\in\mathcal{V}_2} c_q \frac{\mathrm{d}u_q}{\mathrm{d}t}\, w_q = F(w) \qquad \forall\, w \in H^1(\Gamma), \tag{17}$$

where the bilinear form $a : H^1(\Gamma) \times H^1(\Gamma) \to \mathbb{R}$ is given by

$$a(u,w) = \sum_{e\in\mathcal{E}} \int_e \left[\mu_e \frac{\partial u_e}{\partial s}\frac{\partial w_e}{\partial s} + r_e\, u_e\, w_e\right] \mathrm{d}s - \sum_{v_q\in\mathcal{V}_2} r_q\, u_q\, w_q, \tag{18}$$

and the linear functional $F : H^1(\Gamma) \to \mathbb{R}$ is

$$F(w) = \sum_{e\in\mathcal{E}} \int_e f_e\, w_e\, \mathrm{d}s - \sum_{v_q\in\mathcal{V}_2} f_q\, w_q. \tag{19}$$

B.3 Weak formulation

We start from the strong problem on each edge $e \in \mathcal{E}$:

$$\frac{\partial u_e}{\partial t} - \frac{\partial}{\partial s}\left(\mu_e \frac{\partial u_e}{\partial s}\right) + r_e\, u_e = f_e, \qquad s \in (0, \ell_e), \quad t \in (0, T], \tag{B.1}$$

supplemented by the Kirchhoff conditions (14)–(15), the Neumann constraints at the terminal vertices and the global continuity requirements.
Let $V = H^1(\Gamma)$ be the space of functions $w : \Gamma \to \mathbb{R}$ such that $w_e \in H^1(0, \ell_e)$ for each $e \in \mathcal{E}$ and w is continuous at every junction vertex. We multiply (B.1) by a test function $w \in V$, integrate over e and sum over all edges:

$$\sum_{e\in\mathcal{E}} \int_e \frac{\partial u_e}{\partial t}\, w\, \mathrm{d}s - \sum_{e\in\mathcal{E}} \int_e \frac{\partial}{\partial s}\left(\mu_e \frac{\partial u_e}{\partial s}\right) w\, \mathrm{d}s + \sum_{e\in\mathcal{E}} \int_e r_e\, u_e\, w\, \mathrm{d}s = \sum_{e\in\mathcal{E}} \int_e f_e\, w\, \mathrm{d}s. \tag{B.2}$$

We integrate by parts the diffusion term on each edge:

$$-\int_e \frac{\partial}{\partial s}\left(\mu_e \frac{\partial u_e}{\partial s}\right) w\, \mathrm{d}s = \int_e \mu_e \frac{\partial u_e}{\partial s}\frac{\partial w}{\partial s}\, \mathrm{d}s - \left[\mu_e \frac{\partial u_e}{\partial \mathbf{n}_e}\, w\right]_{\partial e}, \tag{B.3}$$

where the boundary term is evaluated at the two endpoints of e with outward normal sign. Summing over all edges and grouping boundary terms by vertex:

$$\sum_{e\in\mathcal{E}} \left[\mu_e \frac{\partial u_e}{\partial \mathbf{n}_e}\, w\right]_{\partial e} = \sum_{v\in\mathcal{V}} w(v) \sum_{e\in\mathcal{E}_v} \left[\mu_e \frac{\partial u_e}{\partial \mathbf{n}_e}\right]_v. \tag{B.4}$$

We now apply the boundary and junction conditions at each vertex:

- At terminal vertices $v \in \mathcal{T}$: if homogeneous Neumann, $[\mu_e \partial_{\mathbf{n}_e} u_e]_v = 0$ and the contribution vanishes. If Dirichlet, $w(v) = 0$ for test functions in the appropriate subspace and again the contribution vanishes.
- At simple junction vertices, the Kirchhoff condition (14) gives $\sum_{e\in\mathcal{E}_{v_p}} [\mu_e \partial_{\mathbf{n}_e} u_e]_{v_p} = 0$, so the contribution vanishes.
- At generalized junction vertices $v_q \in \mathcal{V}_2$: the Kirchhoff condition (15) gives

$$\sum_{e\in\mathcal{E}_{v_q}} \left[\mu_e \frac{\partial u_e}{\partial \mathbf{n}_e}\right]_{v_q} = c_q \frac{\mathrm{d}u_q}{\mathrm{d}t} + r_q\, u_q - f_q.$$

Therefore (B.4) becomes:

$$\sum_{e\in\mathcal{E}} \left[\mu_e \frac{\partial u_e}{\partial \mathbf{n}_e}\, w\right]_{\partial e} = \sum_{v_q\in\mathcal{V}_2} w_q \left(c_q \frac{\mathrm{d}u_q}{\mathrm{d}t} + r_q\, u_q - f_q\right). \tag{B.5}$$

Substituting (B.3) and (B.5) into (B.2) and rearranging:

$$\sum_{e\in\mathcal{E}}\int_e \frac{\partial u_e}{\partial t}\, w \,\mathrm{d}s + a(u,w) - \sum_{v_q\in\mathcal{V}_2} c_q \frac{\mathrm{d}u_q}{\mathrm{d}t}\, w_q = F(w) \qquad \forall\, w\in V, \tag{B.6}$$

where

$$a(u,w) = \sum_{e\in\mathcal{E}}\int_e \left[\mu_e \frac{\partial u_e}{\partial s}\frac{\partial w_e}{\partial s} + r_e\, u_e\, w_e\right]\mathrm{d}s - \sum_{v_q\in\mathcal{V}_2} r_q\, u_q\, w_q,$$

$$F(w) = \sum_{e\in\mathcal{E}}\int_e f_e\, w\,\mathrm{d}s - \sum_{v_q\in\mathcal{V}_2} f_q\, w_q.$$

This is the weak formulation (17).

B.4 Galerkin formulation

Let $V_h \subset V$ be the finite-dimensional subspace spanned by the basis functions $\{\phi_j^e\}_{e\in\mathcal{E},\, j=1,\ldots,n_e-1}$ (interior node hat functions on each edge) and $\{\phi_v\}_{v\in\mathcal{V}}$ (vertex basis functions), satisfying $\phi_j^e(s_k^{e'}) = \delta_{ee'}\delta_{jk}$ and $\phi_v(v') = \delta_{vv'}$. The discrete solution is written as

$$u_h(s,t) = \sum_{e\in\mathcal{E}}\sum_{j=1}^{n_e-1} \alpha_j^e(t)\,\phi_j^e(s) + \sum_{v\in\mathcal{V}} \alpha_v(t)\,\phi_v(s). \tag{B.7}$$

Substituting (B.7) into (B.6) and testing on each basis function, we obtain two families of equations.

Testing on $\phi_k^{e'}$, interior node of edge e'. Since $\phi_k^{e'}$ has support only on e' and vanishes at all vertices, the $\mathcal{V}_2$ term in (B.6) does not contribute. The time term gives:

$$\sum_{e\in\mathcal{E}}\int_e \frac{\partial u_h}{\partial t}\,\phi_k^{e'}\,\mathrm{d}s = \sum_{j=1}^{n_{e'}-1} \dot{\alpha}_j^{e'} \underbrace{\int_{e'} \phi_j^{e'}\,\phi_k^{e'}\,\mathrm{d}s}_{P_{kj}^{e'}} + \sum_{v\in\mathcal{V}} \dot{\alpha}_v \underbrace{\int_{e'} \phi_v\,\phi_k^{e'}\,\mathrm{d}s}_{R_{kv}^{e'}},$$

where only the $e = e'$ term survives due to the local support of $\phi_k^{e'}$ and $R_{kv}^{e'}$ is nonzero only for the (at most two) vertices incident to e'. The stiffness term gives:

$$a(u_h,\phi_k^{e'}) = \sum_{j=1}^{n_{e'}-1} \alpha_j^{e'} \underbrace{\int_{e'}\left[\mu_{e'}\frac{\partial\phi_j^{e'}}{\partial s}\frac{\partial\phi_k^{e'}}{\partial s} + r_{e'}\phi_j^{e'}\,\phi_k^{e'}\right]\mathrm{d}s}_{A_{kj}^{e'}} +$$

$$+ \sum_{v\in\mathcal{V}} \alpha_v \underbrace{\int_{e'}\left[\mu_{e'}\frac{\partial\phi_v}{\partial s}\frac{\partial\phi_k^{e'}}{\partial s} + r_{e'}\phi_v\phi_k^{e'}\right]\mathrm{d}s}_{B_{kv}^{e'}}.$$

Testing on $\phi_{v'}$, vertex basis function. Since $\phi_{v'}(v_q) = \delta_{v'v_q}$, the $\mathcal{V}_2$ time term in (B.6) contributes $-c_{v'}\dot{\alpha}_{v'}$ if $v'\in\mathcal{V}_2$ and zero otherwise. The remaining time term gives:

$$\sum_{e\in\mathcal{E}}\int_e \frac{\partial u_h}{\partial t}\,\phi_{v'}\,\mathrm{d}s = \sum_{e\in\mathcal{E}}\sum_{j=1}^{n_e-1} \dot{\alpha}_j^e \underbrace{\int_e \phi_j^e\phi_{v'}\,\mathrm{d}s}_{R_{v'j}^e} + \sum_{v\in\mathcal{V}} \dot{\alpha}_v \underbrace{\sum_{e\in\mathcal{E}}\int_e \phi_v\phi_{v'}\,\mathrm{d}s}_{Q_{v'v}},$$

where $Q_{vv'}$ is nonzero only when v and v' share at least one edge. The stiffness term gives:

$$a(u_h,\phi_{v'}) = \sum_{e\in\mathcal{E}}\sum_{j=1}^{n_e-1} \alpha_j^e \underbrace{\int_e\left[\mu_e\frac{\partial\phi_j^e}{\partial s}\frac{\partial\phi_{v'}}{\partial s} + r_e\phi_j^e\phi_{v'}\right]\mathrm{d}s}_{B_{v'j}^e} +$$

$$+ \sum_{v\in\mathcal{V}} \alpha_v \underbrace{\left[\sum_{e\in\mathcal{E}}\int_e\left(\mu_e\frac{\partial\phi_v}{\partial s}\frac{\partial\phi_{v'}}{\partial s} + r_e\phi_v\phi_{v'}\right)\mathrm{d}s - r_v\delta_{vv'}\cdot\mathbf{1}_{[v\in\mathcal{V}_2]}\right]}_{G_{v'v}},$$

where $\mathbf{1}_{[v\in\mathcal{V}_2]}$ denotes the indicator function of the set $\mathcal{V}_2$, equal to 1 if v is a junction of the second type and 0 otherwise.

B.5 Matrix formulation

Collecting all unknowns into $\mathbf{u} = [\mathbf{u}_{\mathcal{E}},\, \mathbf{u}_{\mathcal{V}}]^\top$, where $\mathbf{u}_{\mathcal{E}} = \{\alpha_j^e\}$ and $\mathbf{u}_{\mathcal{V}} = \{\alpha_v\}$, the Galerkin system is

$$\mathbf{M}\,\dot{\mathbf{u}}(t) + \mathbf{L}\,\mathbf{u}(t) = \mathbf{f}(t), \tag{B.8}$$

with mass matrix and stiffness matrix

$$\mathbf{M} = \begin{bmatrix} \mathbf{P} & \mathbf{R}^\top \\ \mathbf{R} & \mathbf{Q}+\mathbf{C} \end{bmatrix}, \qquad \mathbf{L} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{G} \end{bmatrix}, \tag{B.9}$$

where the blocks are defined as follows:

- $P_{kj}^e = \displaystyle\int_e \phi_j^e \phi_k^e \,\mathrm{d}s$;
- $R_{kv}^e = \displaystyle\int_e \phi_v \phi_k^e \,\mathrm{d}s$;
- $Q_{v'v} = \displaystyle\sum_{e\in\mathcal{E}} \int_e \phi_v \phi_{v'} \,\mathrm{d}s$;
- $C_{qq} = -c_q$ for $v_q \in \mathcal{V}_2$, zero elsewhere;
- $A_{kj}^e = \displaystyle\int_e \left[\mu_e \frac{\partial \phi_j^e}{\partial s}\frac{\partial \phi_k^e}{\partial s} + r_e \phi_j^e \phi_k^e\right] \mathrm{d}s$;
- $B_{vj}^e = \displaystyle\int_e \left[\mu_e \frac{\partial \phi_j^e}{\partial s}\frac{\partial \phi_v}{\partial s} + r_e \phi_j^e \phi_v\right] \mathrm{d}s$;
- $G_{v'v} = \displaystyle\sum_{e\in\mathcal{E}} \int_e \left[\mu_e \frac{\partial \phi_v}{\partial s}\frac{\partial \phi_{v'}}{\partial s} + r_e \phi_v \phi_{v'}\right] \mathrm{d}s - r_v \delta_{vv'} \cdot \mathbf{1}_{[v\in\mathcal{V}_2]}$.

B.6 Fully discrete system

Discretizing (B.8) in time by the θ-method with uniform step $\Delta t = T/N_T$ and setting $\mathbf{u}^n \approx \mathbf{u}(t^n)$:

$$\mathbf{M}\,\frac{\mathbf{u}^n - \mathbf{u}^{n-1}}{\Delta t} + \mathbf{L}\left(\theta\,\mathbf{u}^n + (1-\theta)\,\mathbf{u}^{n-1}\right) = \theta\,\mathbf{f}^n + (1-\theta)\,\mathbf{f}^{n-1},$$

which rearranges to

$$\mathbf{H}_\theta\,\mathbf{u}^n = (\mathbf{M} - (1-\theta)\Delta t\,\mathbf{L})\,\mathbf{u}^{n-1} + \Delta t\left(\theta\,\mathbf{f}^n + (1-\theta)\,\mathbf{f}^{n-1}\right), \tag{B.10}$$

where $\mathbf{H}_\theta = \mathbf{M} + \theta\,\Delta t\,\mathbf{L}$. The choice $\theta = 1$ recovers the implicit Euler method, while $\theta = 1/2$ gives the Crank-Nicolson scheme. Explicitly:

$$\mathbf{H}_\theta = \begin{bmatrix} \mathbf{P} + \theta\,\Delta t\,\mathbf{A} & \mathbf{R}^\top + \theta\,\Delta t\,\mathbf{B} \\ \mathbf{R} + \theta\,\Delta t\,\mathbf{B}^\top & \mathbf{Q} + \mathbf{C} + \theta\,\Delta t\,\mathbf{G} \end{bmatrix}. \tag{B.11}$$

Since $a(\cdot,\cdot)$ is symmetric and coercive on V and $\mathbf{M}$ is symmetric positive definite, $\mathbf{H}_\theta$ is symmetric positive definite for any $\Delta t > 0$ and $\theta \in (0,1]$ and the algebraic formulation of the parabolic case of partial differential equations on graphs admits a unique solution at each time step.

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